AdaBoost Does Not Always Cycle: A Computer-Assisted Counterexample

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Abstract

We give a computer-assisted counterexample to the open question, posed by Rudin, Schapire, and Daubechies in COLT 2012, of whether exhaustive AdaBoost always converges to a finite cycle. The construction is based on a block-product gadget whose two factors share an exact period-2 orbit for their 5-step branch maps, but whose linearized return maps have dominant eigenvalues with an irrational logarithmic ratio. This irrationality forces the burst-winner sequence to have an irrational asymptotic frequency, precluding eventual periodicity. All assertions are certified by exact rational arithmetic. This work was developed in collaboration with GPT-5.4 Pro and Claude Opus 4.6.

1 Introduction

The AdaBoost algorithm [3] defines a discrete dynamical system on the probability simplex in $\mathbb{R}^m$ [7], where a sequence of distributions D_t is updated based on an $m \times N$ matrix $M \in \{-1, +1\}^{m \times N}$ representing the correlations between N weak hypotheses and m training examples. Under the exhaustive selection of weak hypotheses, the next distribution is determined by the mapping $D_{t+1}(i) = D_t(i)/(1 + r_t M_{ij_t})$, where $j_t = \operatorname{argmax}_j (D_t^\top M)_j$ and $r_t = (D_t^\top M)_{j_t}$. While it is proven that D_t converges to a unique fixed point when the weak learning assumption ($r_t \geq c > 0$) is violated, the behavior remains mathematically elusive when the assumption holds. The iterative map is highly discontinuous due to the argmax operation, leaving the long-term asymptotic behavior—specifically whether the sequence $\{D_t\}$ always enters a cycle of length l such that $D_{kl+b} \to \bar{D}_b$ as $k \to \infty$—an open question [7, 8].

The problem under consideration is the following: prove or disprove that for every $\{-1, +1\}$-valued matrix M, the sequence of distributions D_t computed by the AdaBoost [8] mapping necessarily converges to a cycle. In this paper, we disprove the conjecture by constructing an explicit counterexample.[1]

Proof outline

We construct a 20×8 sign matrix $M_0 = L_A \boxplus L_B$ as a block product of two small gadgets whose dynamics decouple on an invariant product manifold (Section 3). Both gadgets share an exact period-2 orbit for the 5-step branch maps (Section 6). The linearized return maps at this orbit have dominant eigenvalues λ and κ whose ratio $\log \lambda / \log \kappa$ is irrational, as shown by a norm argument in

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[1]The computational certificate is available at this link: `https://github.com/ewang26/AdaBoost`

algebraic number theory (Section 7). Rigorous interval arithmetic certifies explicit invariant sectors around the orbit, together with contraction rates and burst-locking margins (Sections 8–10). Within these sectors, the irrational eigenvalue ratio forces the burst-winner sequence to have an irrational asymptotic frequency, which precludes eventual periodicity and hence finite-cycle convergence (Sections 11–13). A row-duplication argument converts an explicit rational starting point into a uniform-start counterexample on a larger matrix $\widetilde{M}$ with weak-learning margin 1/5 (Section 14).

2 Main result

We work with exhaustive AdaBoost on a fixed sign matrix $M \in \{-1,+1\}^{m\times N}$, with smallest-index tie-breaking. Defining the *edge* $\mu_t(j)$ of hypothesis j at time t by

$$\mu_t(j) = (D_t^\top M)_j, \qquad j_t = \min \operatorname*{argmax}_{1\le j\le N} \mu_t(j), \qquad r_t = \mu_t(j_t),$$

the update is

$$D_{t+1}(i) = \frac{D_t(i)}{1 + r_t M_{ij_t}}.$$

We call the distribution $D_t \in \Delta^{m-1}$ the *state* of the system at time t; the map $D_t \mapsto D_{t+1}$ defined above is a single AdaBoost *update step*. Let Δ^{m-1} be the $m-1$-dimensional simplex in $\mathbb{R}^m$. The *weak-learning margin* of M is

$$\gamma(M) = \min_{D\in\Delta^{m-1}} \max_{1\le j\le N} (D^\top M)_j.$$

We show the following result:

Theorem 1. *There exists a finite $\{-1,+1\}$-valued matrix $\widetilde{M}$ with weak-learning margin $\gamma(\widetilde{M}) = \frac{1}{5} > 0$ such that standard exhaustive AdaBoost on $\widetilde{M}$, started from the exact uniform distribution and using smallest-index tie-breaking, does **not** converge to any finite cycle.*

*Consequently, the COLT question "does exhaustive AdaBoost always converge to a finite cycle?" has a **negative** answer.*

Corollary 2. *There exists a finite $\{-1,+1\}$-valued matrix $\widetilde{M}'$ whose column set is closed under negation (i.e., for every column h of $\widetilde{M}'$, $-h$ also appears as a column of $\widetilde{M}'$), with weak-learning margin $\gamma(\widetilde{M}') = \frac{1}{5} > 0$, such that standard exhaustive AdaBoost on $\widetilde{M}'$, started from the exact uniform distribution and using smallest-index tie-breaking, does **not** converge to any finite cycle. In particular, the COLT question in its original hypothesis-class-closed-under-negation formulation also has a negative answer.*

The proof of the Corollary appears at the end of Section 14.

The remainder of the paper is devoted to the proof of Theorem 1, which occupies Sections 3–14.

Our work comes with an accompanying symbolic certificate using only symbolic algebra over $\mathbb{Q}$ (via [5]), exact interval arithmetic, and exact square-root enclosures obtained by integer arithmetic. In the presentation of this paper, decimal intervals are used only for reporting.

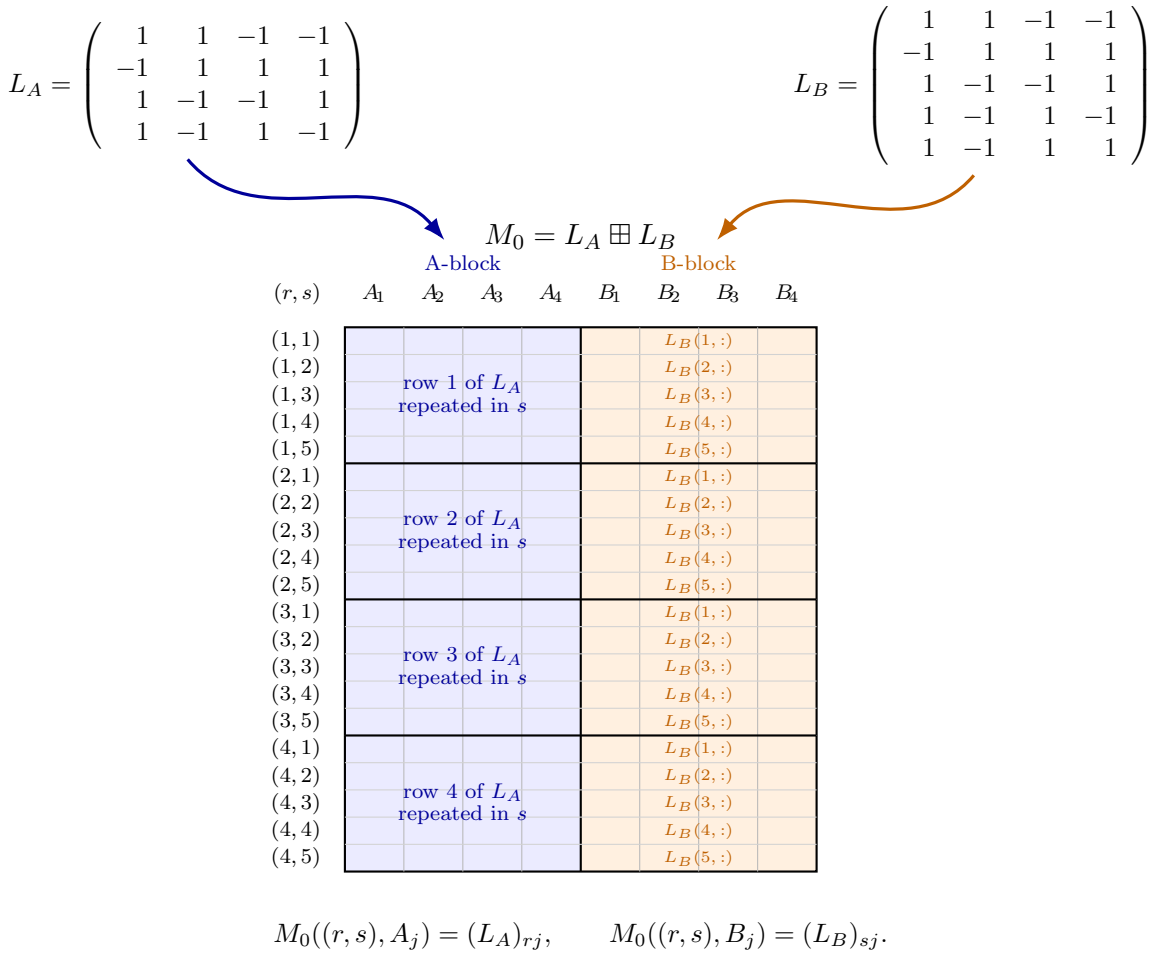


Figure 1: Schematic construction of $M_0 = L_A \boxplus L_B$. Rows are indexed by $(r,s) \in \{1,2,3,4\} \times \{1,2,3,4,5\}$. The first four columns are copied from row r of L_A, while the last four columns are copied from row s of L_B.

3 The gadgets, the product system, and the exact invariant affine manifolds

The counterexample is built from two small sign matrices, which we call *gadgets*. Let

$$L_A = \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad L_B = \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}.$$

Define the 20×8 block-product matrix $M_0 = L_A \boxplus L_B$ as follows (see Figure 1). Its rows are indexed by pairs $(r,s) \in \mathcal{I} := \{1,2,3,4\} \times \{1,2,3,4,5\}$, and its columns are denoted $A_1, \ldots, A_4, B_1, \ldots, B_4$. The entries are

$$M_0\big((r,s), A_j\big) = (L_A)_{rj}, \qquad M_0\big((r,s), B_j\big) = (L_B)_{sj}.$$

Lemma 3 (Reduction to row marginals)**.** *Let $W = (w_{rs})_{(r,s)\in\mathcal{I}}$ be any distribution on the* 20 *rows of M_0, and define its row marginals*

$$a_r = \sum_{s=1}^{5} w_{rs}, \qquad b_s = \sum_{r=1}^{4} w_{rs}.$$

Then for the edge vector $W^\top M_0 \in \mathbb{R}^8$*, whose jth component $(W^\top M_0)_j = \sum_{(r,s)} w_{rs}(M_0)_{(r,s),j}$ is the inner product of W with column j, the A-block edges depend only on a, and the B-block edges depend only on b:*

$$\begin{aligned}(W^\top M_0)_{A_j} &= (a^\top L_A)_j \qquad (1 \le j \le 4),\\ (W^\top M_0)_{B_j} &= (b^\top L_B)_j \qquad (1 \le j \le 4).\end{aligned}$$

Proof. For $1 \le j \le 4$,

$$(W^\top M_0)_{A_j} = \sum_{r=1}^{4}\sum_{s=1}^{5} w_{rs}(L_A)_{rj} = \sum_{r=1}^{4} a_r(L_A)_{rj} = (a^\top L_A)_j.$$

The B-block identity is similar. □

Lemma 4 (Exact invariance of the product manifold)**.** *Let*

$$\mathcal{P} = \{a \otimes b : \ a \in \Delta^3, \ b \in \Delta^4\}.$$

If $W = a \otimes b \in \mathcal{P}$ and AdaBoost selects an A-column A_j, then the updated state is $W' = a' \otimes b$, where a' is the one-step exhaustive-AdaBoost update of a on L_A using column j. If AdaBoost selects a B-column B_j, then $W' = a \otimes b'$, where b' is the one-step update of b on L_B using column j. Hence $\mathcal{P}$ is invariant *under the AdaBoost dynamics on M_0: if $W \in \mathcal{P}$, then the updated state $W' \in \mathcal{P}$.*

Proof. Assume an A-column j is selected. By Lemma 3, the selected edge is $\mu = (a^\top L_A)_j$. Therefore

$$w'_{rs} = \frac{a_r b_s}{1 + \mu \cdot (L_A)_{rj}} = \left(\frac{a_r}{1 + \mu \cdot (L_A)_{rj}}\right) b_s = a'_r b_s,$$

where a' is the one-step AdaBoost update of a on L_A. The B-column case is identical. □

We will show in Section 10 that the product dynamics over $\mathcal{P}$ decouples into alternating blocks of five consecutive AdaBoost steps in which the same gadget is selected at every step. We call each such block of five steps a *burst*. Within a burst, the column-selection sequence on the active gadget is one of two possible 5-tuples, called *half-words*,

$$H_0 = (1, 3, 4, 1, 2), \qquad H_1 = (1, 4, 3, 1, 2).$$

The choice between these is determined by the *phase*: if the active gadget's current state is near the first point $\hat{P}_0^G$ of a certain 2-cycle (Section 6), we say it is in phase 0; if near $\hat{P}_1^G$, in phase 1.

Since both H_0 and H_1 end in column 2, and column 2 of L_A is $(1, 1, -1, -1)^\top$, the classical balancing lemma for AdaBoost (see [9] and Section A.1) implies that after the final step of any A-burst, the total mass on the $+1$ rows (rows $1, 2$) and on the -1 rows (rows $3, 4$) is each exactly $\frac{1}{2}$. Writing the distribution as $(a, b, c, d) \in \Delta^3$, this means $a + b = \frac{1}{2}$ and $c + d = \frac{1}{2}$. Hence every 5-step A-burst lands in

$$\Sigma_A = \{(a, b, c, d) \in \Delta^3 : \ a + b = \tfrac{1}{2}, \ c + d = \tfrac{1}{2}\}.$$

Similarly, column 2 of L_B is $(1,1,-1,-1,-1)^\top$, so Lemma 26 gives $a+b=\frac{1}{2}$ and $c+d+e=\frac{1}{2}$. Every 5-step B-burst lands in

$$\Sigma_B = \{(a,b,c,d,e) \in \Delta^4 : \ a+b=\tfrac{1}{2},\ c+d+e=\tfrac{1}{2}\}.$$

Thus, for some p, d, we can represent

$$(a,b,c,d) = \Big(p-\tfrac{1}{2},\ 1-p,\ \tfrac{1}{2}-d,\ d\Big) \in \Sigma_A,$$
$$(a,b,c,d,e) = \Big(p-\tfrac{1}{2},\ 1-p,\ \tfrac{1}{2}-d-e,\ d,\ e\Big) \in \Sigma_B.$$

For each gadget $G \in \{A,B\}$ and each half-word H_i, composing the five consecutive AdaBoost update steps prescribed by H_i yields a rational map $\Sigma_G \to \Sigma_G$ called the *branch map* F_i^G (explicit formulas are given in Section 5). Item 2 of the certificate (Appendix A.2) verifies symbolically that each numerator polynomial of the last coordinate of F_i^B is divisible, as a polynomial in $\mathbb{Q}[p,d,e]$, by the variable e. Consequently, the *face* $\{e=0\} \subset \Sigma_B$—i.e., the subset of Σ_B obtained by setting the fifth simplicial coordinate to zero—is invariant under both branch maps, and restriction to this face recovers the A-gadget maps. Concretely, the following structural identity holds (which Item 2 of the certificate also verifies symbolically).

Let U_j^G denote the one-step exhaustive AdaBoost update on L_G using column j, and for a finite word $w=(j_1,\dots,j_k)$ in the four columns, define the composition $T_w^G = U_{j_k}^G \circ \cdots \circ U_{j_1}^G$.

Lemma 5 (Zero-mass face and restriction of L_B to L_A)**.** *Let*

$$\iota : \Delta^3 \to \Delta^4, \qquad \iota(a_1,a_2,a_3,a_4) = (a_1,a_2,a_3,a_4,0).$$

For every finite word w in the four columns and every $a \in \Delta^3$ for which $T_w^A(a)$ is well-defined—that is, for which every denominator $1+\mu\cdot(L_A)_{rj}$ encountered along the word is nonzero (in particular, for every strictly positive a, i.e., a in the open simplex)—we have

$$T_w^B \circ \iota(a) = \iota \circ T_w^A(a),$$

and the right-hand side determines a well-defined $T_w^B(\iota(a))$ via the same denominator condition. Consequently, in the affine charts

$$a(p,d) = \big(p-\tfrac{1}{2},\ 1-p,\ \tfrac{1}{2}-d,\ d\big),$$

$$b(p,d,e) = \big(p-\tfrac{1}{2},\ 1-p,\ \tfrac{1}{2}-d-e,\ d,\ e\big),$$

the branch maps satisfy, on the regular domain Σ_A° of Section 5,

$$F_i^B(p,d,0) = \big(F_i^A(p,d),0\big).$$

In particular,

$$F_0^B|_{e=0} = F_0^A, \qquad F_1^B|_{e=0} = F_1^A.$$

Proof. It suffices to prove the one-step claim under the regularity hypothesis (the multi-step case follows by induction on w, where regularity of each prefix is the inductive hypothesis applied at each step). Start L_B from $\iota(a)$. For column j,

$$\mu_B = \sum_{r=1}^{5} \iota(a)_r (L_B)_{rj} = \sum_{r=1}^{4} a_r (L_A)_{rj} = \mu_A,$$

because the first four rows of L_B equal L_A, and the fifth coordinate has mass zero.

For $1 \le r \le 4$,

$$\frac{\iota(a)_r}{1+\mu_B(L_B)_{rj}} = \frac{a_r}{1+\mu_A(L_A)_{rj}},$$

which is the r-th coordinate of the L_A update. The fifth coordinate remains

$$\frac{0}{1+\mu_B(L_B)_{5j}} = 0.$$

Induction over the word w proves the word-map identity. The chart identity is exactly this word-map identity expressed in (p, d, e) coordinates. □

Notation convention. Throughout, we distinguish *affine chart points* from *simplex probability vectors* using a hat. We write

$$\hat{P}_i^A = (p_i, d_i) \in \mathbb{R}^2, \qquad \hat{P}_i^B = (p_i, d_i, e_i) \in \mathbb{R}^3$$

for chart coordinates on Σ_A and Σ_B, and $P_i^A \in \Delta^3$, $P_i^B \in \Delta^4$ for the corresponding simplex probability vectors obtained from the affine parametrizations above. Similarly, at burst boundary n (i.e., the state immediately before the nth burst begins), we write $\hat{a}_n, \hat{b}_n$ for chart coordinates and a_n, b_n for the simplex vectors. We define $W_n = a_n \otimes b_n \in \mathcal{P}$ to be the product distribution on the 20 rows of M_0 at the nth burst boundary.

4 Margin

We show using a single linear witness that the weak-learning margin of the block-product matrix M_0 is exactly $1/5$. Let $\beta = (2, 1, 1, 1)^\top \in \mathbb{R}^{1\times 4}$. Every row r of L_A satisfies $r\beta = 1$, and every row r of L_B satisfies $r\beta \in \{1, 3\}$. For any gadget $G \in \{A, B\}$ and any distribution u on the rows of L_G, write $\mu_j = (u^\top L_G)_j$ for the edge of column j. Then

$$2\mu_1 + \mu_2 + \mu_3 + \mu_4 = u^\top L_G\, \beta \ge 1,$$

so $\max_j \mu_j \ge 1/5$. Hence $\gamma(L_A), \gamma(L_B) \ge \frac{1}{5}$.

Equality for L_A is attained at

$$u_* = (1/5, 2/5, 1/5, 1/5), \qquad u_*^\top L_A = (1/5, 1/5, 1/5, 1/5).$$

It follows that $\gamma(L_A) = 1/5$. Since the first four rows of L_B coincide with L_A, the same argument gives $\gamma(L_B) \le 1/5$, and together with the lower bound above, $\gamma(L_A) = \gamma(L_B) = \frac{1}{5}$.

By Lemma 3, for any distribution W on the rows of M_0 with marginals (a, b),

$$\max_{1\le j\le 8} (W^\top M_0)_j = \max\Big(\max_{1\le j\le 4} (a^\top L_A)_j,\ \max_{1\le j\le 4} (b^\top L_B)_j\Big).$$

Conversely, every pair of marginals (a, b) is realized by the product distribution $a \otimes b$. Hence

$$\gamma(M_0) = \min_W \max_j (W^\top M_0)_j = \min_{a,b} \max\Big(\max_j (a^\top L_A)_j,\ \max_j (b^\top L_B)_j\Big) = \frac{1}{5}.$$

We now formalize the conclusions of the preceding discussion as a general block-product margin identity, followed by the concrete margin computation for L_A, L_B, and M_0.

Lemma 6 (Block-product margin)**.** *Let $C = A \boxplus B$. Then*

$$\gamma(C) = \max(\gamma(A), \gamma(B)).$$

Proof. For any distribution W on product rows, let a, b be its marginals. By Lemma 3,

$$\max_j (W^\top C)_j = \max\Big\{\max_j (a^\top A)_j,\ \max_k (b^\top B)_k\Big\}.$$

Since

$$\max_j (a^\top A)_j \geq \gamma(A), \qquad \max_k (b^\top B)_k \geq \gamma(B),$$

we get

$$\max_j (W^\top C)_j \geq \max(\gamma(A), \gamma(B)).$$

Taking the minimum over W,

$$\gamma(C) \geq \max(\gamma(A), \gamma(B)).$$

Conversely, because the simplices are compact and the objective functions are continuous, choose minimizers a_*, b_* for A and B. Let

$$W_* = a_* \otimes b_*.$$

Then

$$\max_j (W_*^\top C)_j = \max(\gamma(A), \gamma(B)).$$

Hence

$$\gamma(C) \leq \max(\gamma(A), \gamma(B)).$$

Therefore equality holds. □

Proposition 7 (Concrete margin of L_A, L_B, M_0)**.** *For the concrete matrices L_A, L_B,*

$$\gamma(L_A) = \gamma(L_B) = \frac{1}{5}.$$

Therefore, for

$$M_0 = L_A \boxplus L_B,$$
$$\gamma(M_0) = \frac{1}{5}.$$

Proof. Let

$$\beta = (2, 1, 1, 1)^\top.$$

For L_A, every row r satisfies

$$r\beta = 1.$$

Indeed,

$$(1, 1, -1, -1)\beta = 2 + 1 - 1 - 1 = 1,$$
$$(-1, 1, 1, 1)\beta = -2 + 1 + 1 + 1 = 1,$$
$$(1, -1, -1, 1)\beta = 2 - 1 - 1 + 1 = 1,$$
$$(1, -1, 1, -1)\beta = 2 - 1 + 1 - 1 = 1.$$

For L_B, the first four rows have the same dot products, and the fifth row satisfies

$$(1,-1,1,1)\beta = 2-1+1+1 = 3.$$

Let $q = \beta/5$. Then q is a column distribution. For L_A,

$$(L_A q)_r = \frac{1}{5}$$

for every row. For L_B,

$$(L_B q)_r \geq \frac{1}{5}$$

for every row. By the minimax theorem,

$$\gamma(L_A) \geq \frac{1}{5}, \qquad \gamma(L_B) \geq \frac{1}{5}.$$

For the reverse inequalities, take row distributions

$$u_A = \left(\tfrac{1}{5}, \tfrac{2}{5}, \tfrac{1}{5}, \tfrac{1}{5}\right),$$
$$u_B = \left(\tfrac{1}{5}, \tfrac{2}{5}, \tfrac{1}{5}, \tfrac{1}{5}, 0\right).$$

Direct multiplication gives

$$u_A^\top L_A = \left(\tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{5}\right),$$
$$u_B^\top L_B = \left(\tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{5}\right).$$

Hence

$$\gamma(L_A) \leq \frac{1}{5}, \qquad \gamma(L_B) \leq \frac{1}{5}.$$

Thus

$$\gamma(L_A) = \gamma(L_B) = \frac{1}{5}.$$

By Lemma 6,

$$\gamma(M_0) = \max(\gamma(L_A), \gamma(L_B)) = \frac{1}{5}.$$

□

5 Exact 5-step branch maps

We now compute the closed-form rational maps that result from composing five consecutive AdaBoost updates along each half-word. The two branch maps F_0^G and F_1^G for each gadget $G \in \{A, B\}$ are related by a simple involution. Define the *phase-swap* maps

$$S_A(p,d) = \left(p, \tfrac{1}{2} - d\right), \qquad S_B(p,d,e) = \left(p, \tfrac{1}{2} - d - e,\ e\right).$$

Each S_G is an involution ($S_G^2 = \mathrm{id}$). Using simple algebra, one can check that in simplex coordinates, both maps S_G *transpose the 3rd and 4th distribution entries* while leaving the rest fixed.

For either gadget, rows 3 and 4 of L_G differ only in columns 3 and 4 (where their signs are opposite), so this row transposition induces the column transposition $\pi = (3\ 4)$ on the edge vector. If T_j^G denotes the one-step AdaBoost update using column j of L_G, then $S_G \circ T_j^G \circ S_G = T_{\pi(j)}^G$. Because $S_G^2 = \mathrm{id}$, conjugating the five-step composition F_0^G by S_G replaces each column index j with $\pi(j)$, transforming $H_0 = (1,3,4,1,2)$ into $H_1 = (1,4,3,1,2)$. Hence,

$$F_1^A = S_A \circ F_0^A \circ S_A, \qquad F_1^B = S_B \circ F_0^B \circ S_B.$$

It therefore suffices to give explicit formulas for F_0^A and F_0^B.

A-gadget

To obtain F_0^A, we substitute the chart parametrization $(p-\frac{1}{2},\, 1-p,\, \frac{1}{2}-d,\, d) \in \Sigma_A$ into the five successive AdaBoost update steps $a_i \mapsto a_i/(1+\mu\, h_i)$ for the columns of L_A prescribed by $H_0 = (1,3,4,1,2)$, then express the final result in the chart coordinates (p,d) and reduce to lowest terms. The phase-0 branch map on Σ_A is the rational function F_0^A such that for all chart coordinates p, d,

$$F_0^A(p,d) = \left(\frac{P_A(p,d)}{Q_A(p,d)}, \frac{D_A(p,d)}{R_A(p,d)}\right),$$

where

$$P_A = 4d^4 - 6d^3p + 2d^3 - 30d^2p^2 + 15d^2p - 2d^2 - 6dp^3 + 14dp^2 - 4dp + 6p^4 + p^3 - 2p^2,$$

$$Q_A = 8d^4 - 8d^3p + 2d^3 - 48d^2p^2 + 22d^2p - 3d^2 - 8dp^3 + 22dp^2 - 6dp + 8p^4 + 2p^3 - 3p^2,$$

$$D_A = d\big(2d^3 + 2d^2p - d^2 - 6dp^2 + 2p^3 + p^2\big),$$

and

$$R_A = 8d^4 - 2d^3 - 24d^2p^2 + 6d^2p - d^2 + 10dp^2 - 2dp + 2p^3 - p^2.$$

B-gadget

The same five-step composition applied to L_B along H_0, in the chart coordinates (p,d,e) on Σ_B, gives the phase-0 branch map on Σ_B as

$$F_0^B(p,d,e) = \left(\frac{P_B(p,d,e)}{Q_B(p,d,e)}, \frac{D_B(p,d,e)}{R_B(p,d,e)}, \frac{E_B(p,d,e)}{R_B(p,d,e)}\right),$$

where

$$\begin{aligned} P_B = {} & 4d^4 + 16d^3e - 6d^3p + 2d^3 + 20d^2e^2 - 36d^2ep + 12d^2e - 30d^2p^2 + 15d^2p - 2d^2 \\ & + 8de^3 - 46de^2p + 18de^2 - 48dep^2 + 30dep - 4de - 6dp^3 + 14dp^2 - 4dp \\ & - 16e^3p + 8e^3 - 22e^2p^2 + 15e^2p - 2e^2 + 8ep^2 - 4ep + 6p^4 + p^3 - 2p^2, \end{aligned}$$

$$\begin{aligned} Q_B = {} & 8d^4 + 32d^3e - 8d^3p + 2d^3 + 40d^2e^2 - 56d^2ep + 16d^2e - 48d^2p^2 + 22d^2p - 3d^2 \\ & + 16de^3 - 72de^2p + 26de^2 - 72dep^2 + 44dep - 6de - 8dp^3 + 22dp^2 - 6dp \\ & - 24e^3p + 12e^3 - 32e^2p^2 + 22e^2p - 3e^2 + 12ep^2 - 6ep + 8p^4 + 2p^3 - 3p^2, \end{aligned}$$

$$D_B = d\big(2d^3 + 8d^2e + 2d^2p - d^2 + 10de^2 - 2de - 6dp^2 + 4e^3 - 2e^2p - e^2 - 4ep^2 + 2p^3 + p^2\big),$$

$$\begin{aligned} R_B = {} & 8d^4 + 32d^3e - 2d^3 + 40d^2e^2 - 24d^2ep - 24d^2p^2 + 6d^2p - d^2 \\ & + 16de^3 - 32de^2p + 6de^2 - 24dep^2 + 12dep - 2de + 10dp^2 - 2dp \\ & - 8e^3p + 4e^3 - 8e^2p^2 + 6e^2p - e^2 + 4ep^2 - 2ep + 2p^3 - p^2, \end{aligned}$$

and

$$E_B = e\big(2d^3 + 4d^2e - 6d^2p + d^2 + 2de^2 - 8dep + 2de + 2dp^2 - 2e^2p + e^2 + 2p^3 - p^2\big).$$

Item 1 of the certificate (Appendix A.2) verifies symbolically that the displayed polynomials for both gadgets match the five-step compositions term-for-term. Because E_B has a factor e, the face $e = 0$ is invariant.

Lemma 8 (Regularity of the branch maps)**.** *The denominators Q_A, R_A are nonzero on $\Sigma_A^\circ := \{(p,d) : \frac{1}{2} < p < 1,\ 0 < d < \frac{1}{2}\}$, and Q_B, R_B are nonzero on $\Sigma_B^\circ = \{(p,d,e) : \frac{1}{2} < p < 1,\ d > 0,\ e > 0,\ d+e < \frac{1}{2}\}$. Moreover, the branch maps F_i^A, F_i^B and all their derivatives are smooth on these domains.*

Proof. Observe that every column of L_A contains both $+1$ and -1 entries.

Strict positivity implies $|\mu| < 1$. If $a \in \Delta^{m-1}$ is strictly positive and $h \in \{-1,+1\}^m$ has both signs, then

$$\mu = a^\top h = \sum_{h_i=+1} a_i - \sum_{h_i=-1} a_i.$$

Both partial sums are strictly positive, so $-1 < \mu < 1$. Hence every denominator $1 + \mu h_i \in \{1+\mu,\, 1-\mu\}$ is strictly positive, and the updated vector $a_i' = a_i/(1+\mu h_i)$ is again strictly positive with $\sum_i a_i' = 1$.

Well-definedness of the five-step composition. A point $(p,d) \in \Sigma_A^\circ$ corresponds to the strictly positive distribution $(p - \frac{1}{2},\, 1-p,\, \frac{1}{2} - d,\, d) \in \Delta^3$. By the above reasoning, each of the five individual AdaBoost updates in the column sequence H_0 (or H_1) has strictly positive denominators and produces a strictly positive output distribution. Hence the five-step composition is well-defined on all of Σ_A°. An identical argument applies to L_B on Σ_B°.

Reduced denominators are nonzero. Before algebraic simplification, each coordinate of the five-step composition is a rational function $\tilde{P}/\tilde{Q}$ whose denominator $\tilde{Q}$ is a product of the individual-step denominators (polynomials in the chart coordinates obtained by successive substitution and clearing of fractions). By the above reasoning, each factor is strictly positive on Σ_A°, so $\tilde{Q} > 0$ there. The reduced-form denominator Q satisfies $Q \mid \tilde{Q}$: since $P\tilde{Q} = Q\tilde{P}$ and $\gcd(P,Q) = 1$, Euclid's lemma in the polynomial ring $\mathbb{Q}[p,d]$ gives $Q \mid \tilde{Q}$. Hence any real zero of Q would also be a zero of $\tilde{Q}$. Since $\tilde{Q}$ has no zeros on Σ_A°, neither does Q. The same argument applies to R_A, Q_B, and R_B.

Since F_i^A and F_i^B are rational functions with nonzero denominators, they are C^∞ on Σ_A° and Σ_B° respectively; finishing the proof. □

6 Exact common 2-cycle

We now argue that both gadgets share an exact period-2 orbit for the 5-step branch maps. By the conjugacy $F_1^A = S_A \circ F_0^A \circ S_A$ (Section 5), if F_0^A maps a point (p,d) to its phase-swap $S_A(p,d) = (p, \frac{1}{2} - d)$, then F_1^A maps the phase-swap back: $F_1^A(p, \frac{1}{2} - d) = S_A(F_0^A(p,d)) = (p,d)$, giving a 2-cycle. We therefore seek solutions of

$$F_0^A(p,d) = \big(p, \tfrac{1}{2} - d\big).$$

Substituting the explicit rational formulas for F_0^A (Section 5), the two coordinate equations $P_A/Q_A = p$ and $D_A/R_A = \frac{1}{2} - d$ become, after cross-multiplying and removing factors that are nonzero on Σ_A°, the polynomial system $N_1(p,d) = 0$, $N_2(p,d) = 0$, where

$$N_1(p,d) = 2d^4 - 2d^3p + d^3 - 12d^2p^2 + 7d^2p - d^2 - 2dp^3 + 6dp^2 - 2dp + 2p^4 - p^2,$$

and

$$\begin{aligned} N_2(p,d) = &-16d^5 + 8d^4 + 48d^3p^2 - 16d^3p + 2d^3 \\ &- 32d^2p^2 + 10d^2p - d^2 - 8dp^3 + 10dp^2 - 2dp + 2p^3 - p^2. \end{aligned}$$

To eliminate d from the system $N_1 = N_2 = 0$, we compute the resultant of N_1 and N_2 with respect to d. By standard elimination theory [1], any common root (p, d) must satisfy $\mathrm{Res}_d(N_1, N_2)(p) = 0$. Item 3 of the certificate (Appendix A.2) verifies that

$$\mathrm{Res}_d(N_1, N_2) = 4096\, p^{10}(2p-1)^3 f(p),$$

where

$$f(p) = 1024p^7 - 3072p^6 + 3200p^5 - 1152p^4 - 144p^3 + 176p^2 - 40p + 7.$$

The factors $p^{10}(2p-1)^3$ are nonzero on Σ_A° (since $p > \frac{1}{2}$), so any interior solution must satisfy $f(p) = 0$.

The certificate (Item 3 of Appendix A.2) isolates a unique root

$$p_* \in [0.637116837818846,\ 0.637116837818848],$$

and proves

$$\begin{aligned} d_* &= 512p_*^6 - 1152p_*^5 + 736p_*^4 - 24p_*^3 - 88p_*^2 + 19p_* - \tfrac{9}{2} \\ &\in [0.257995382953758869,\ 0.257995382958145447]. \end{aligned}$$

In chart coordinates, set

$$\hat{P}_0^A = (p_*,\, d_*), \qquad \hat{P}_1^A = (p_*,\, \tfrac{1}{2} - d_*).$$

The corresponding simplex vectors are

$$P_0^A = \Big(p_* - \tfrac{1}{2},\ 1 - p_*,\ \tfrac{1}{2} - d_*,\ d_*\Big), \qquad P_1^A = \Big(p_* - \tfrac{1}{2},\ 1 - p_*,\ d_*,\ \tfrac{1}{2} - d_*\Big). \tag{6.1}$$

Then $F_0^A(\hat{P}_0^A) = \hat{P}_1^A$, $F_1^A(\hat{P}_1^A) = \hat{P}_0^A$. In chart coordinates, set

$$\hat{P}_0^B = (p_*,\, d_*,\, 0), \qquad \hat{P}_1^B = (p_*,\, \tfrac{1}{2} - d_*,\, 0),$$

with corresponding simplex vectors

$$P_0^B = (P_0^A, 0), \qquad P_1^B = (P_1^A, 0). \tag{6.2}$$

Since $F_i^B|_{e=0} = F_i^A$, the same identities hold for B: $F_0^B(\hat{P}_0^B) = \hat{P}_1^B$, $F_1^B(\hat{P}_1^B) = \hat{P}_0^B$.

Thus both gadgets share an exact period-2 orbit for the 5-step branch maps, equivalently a period-10 orbit in AdaBoost time. We say gadget G is in *phase* i ($i \in \{0, 1\}$) when its state lies in the sector S_i^G (Section 9); within these sectors, the column-selection sequence is exactly H_i (Section 10), so the 5-step composition is the branch map F_i^G.

7 Exact spectra and the irrationality input

Denote by $J_i^G = DF_i^G(\hat{P}_i^G)$ the Jacobian of the branch map F_i^G at the 2-cycle point $\hat{P}_i^G$, and define the *return maps*

$$R_A = J_1^A J_0^A, \qquad R_B = J_1^B J_0^B.$$

Each R_G is the linearization of one full period (phase $0 \to 1 \to 0$) of the 2-cycle for gadget G. We call the binary sequence recording which gadget (A or B) is selected at each successive burst the *burst-winner sequence* (defined precisely in Section 11).

The key algebraic input of this section is that the dominant eigenvalues of R_A and R_B have an irrational logarithmic ratio, which will ultimately force the burst-winner sequence to be aperiodic.

Next, we observe that the return map R_B has a block-triangular structure.

Lemma 9 (Block-triangular B-return map)**.** *Let P_0, P_1 be the common two-cycle points on the face $e = 0$. Let*

$$J_i^A = DF_i^A(P_i), \qquad J_i^B = DF_i^B(P_i, 0),$$

and

$$R_A = J_1^A J_0^A, \qquad R_B = J_1^B J_0^B.$$

Then

$$R_B = \begin{pmatrix} R_A & u \\ 0 & \kappa \end{pmatrix}$$

for some column vector u and scalar κ.

Proof. By Lemma 5,

$$F_i^B(p, d, 0) = \big(F_i^A(p, d), 0\big).$$

Therefore the upper-left 2×2 block of J_i^B is J_i^A.

It remains to prove the lower-left block is zero. In one L_B-update, the fifth coordinate has the form

$$e' = \frac{e}{1 + \mu\sigma}, \qquad \sigma \in \{\pm 1\},$$

where μ is the selected edge. Along any fixed finite word, repeated substitution gives

$$e_{\text{out}} = e\, G_i(p, d, e)$$

for a rational function G_i regular at the two-cycle point. Hence

$$\frac{\partial e_{\text{out}}}{\partial p}(P_i, 0) = 0, \qquad \frac{\partial e_{\text{out}}}{\partial d}(P_i, 0) = 0.$$

Thus each J_i^B has block form

$$J_i^B = \begin{pmatrix} J_i^A & v_i \\ 0 & \eta_i \end{pmatrix}.$$

Multiplying $J_1^B J_0^B$ gives

$$R_B = \begin{pmatrix} J_1^A J_0^A & J_1^A v_0 + v_1 \eta_0 \\ 0 & \eta_1 \eta_0 \end{pmatrix} = \begin{pmatrix} R_A & u \\ 0 & \kappa \end{pmatrix}.$$

□

The dominant eigenvalue of R_A is an algebraic number λ whose irreducible polynomial over $\mathbb{Q}$ is needed explicitly to compute the field norm for the irrationality argument below. It turns out that this value λ is the unique root of the irreducible degree-14 polynomial

$$\begin{aligned} P_\lambda(x) = {} & 16x^{14} - 117232x^{13} - 8330040x^{12} - 3628852593x^{11} \\ & + 766183444916x^{10} - 37440753370248x^9 \\ & + 504085851461616x^8 + 57085439654130x^7 \\ & + 18299281845941356x^6 - 5676298472505780x^5 \\ & + 422743039128068x^4 - 4789675432569x^3 \\ & + 7468158864x^2 - 2899292x + 4 \end{aligned}$$

in the *isolating interval* $[0.0944920594891271,\ 0.0944920594891272]$. The certificate derives P_λ as the elimination polynomial for λ and confirms it is irreducible (Item 5, Appendix A.2), so P_λ is the minimal polynomial of λ over $\mathbb{Q}$. The isolating interval contains exactly one real root of P_λ (Item 7), thereby identifying λ uniquely among the roots.

If α is an algebraic number with irreducible polynomial $a_d x^d + \cdots + a_0$ over $\mathbb{Q}$, its *field norm* $N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha)$ is the product of all roots of this polynomial, which equals $(-1)^d a_0/a_d$ by Vieta's formulas [4]. Hence

$$N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda) = \frac{4}{16} = \frac{1}{4}.$$

The other eigenvalue of R_A is also a root of P_λ.

Similarly, the eigenvalue κ of R_B not inherited from R_A turns out to be the unique root of the irreducible degree-7 polynomial

$$P_\kappa(x) = x^7 - 35x^6 + 385x^5 - 1835x^4 + 2315x^3 - 625x^2 + 51x - 1$$

in the isolating interval $[0.2038817328642625,\ 0.2038817328642626]$. As above, the certificate derives P_κ as the irreducible elimination polynomial for κ (Item 6), making it the minimal polynomial of κ over $\mathbb{Q}$, and verifies that this interval contains exactly one real root (Item 7). Hence $N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa) = 1$.

Defining ν as the unique root of P_λ in $[0.0011453882842312,\ 0.0011453882842313]$ (Item 7 of the certificate), we then find that the isolating intervals give $0 < \nu < \lambda < \kappa < 1$.

The irrationality of the logarithmic ratio now follows from a general norm criterion.

Lemma 10 (Norm criterion for irrational logarithmic ratio)**.** *Let $\lambda, \kappa \in (0,1)$ be algebraic numbers satisfying*

$$N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda) = \frac{1}{4}, \qquad N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa) = 1.$$

Then

$$\frac{\log \lambda}{\log \kappa} \notin \mathbb{Q}.$$

Proof. Suppose for contradiction that

$$\frac{\log \lambda}{\log \kappa} \in \mathbb{Q}.$$

Since $\lambda, \kappa \in (0,1)$, both logarithms are negative, so the ratio is positive. Thus there exist positive integers m, n such that

$$n \log \lambda = m \log \kappa,$$

hence

$$\lambda^n = \kappa^m.$$

Let

$$L = \mathbb{Q}(\lambda, \kappa).$$

Apply the field norm $N_{L/\mathbb{Q}}$ to both sides:

$$N_{L/\mathbb{Q}}(\lambda)^n = N_{L/\mathbb{Q}}(\kappa)^m.$$

By the tower formula,

$$N_{L/\mathbb{Q}}(\lambda) = N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda)^{[L:\mathbb{Q}(\lambda)]} = \left(\frac{1}{4}\right)^{[L:\mathbb{Q}(\lambda)]},$$

whereas

$$N_{L/\mathbb{Q}}(\kappa) = N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa)^{[L:\mathbb{Q}(\kappa)]} = 1.$$

Therefore the norm equality becomes

$$\left(\frac{1}{4}\right)^{n[L:\mathbb{Q}(\lambda)]} = 1,$$

which is impossible. Hence $\log\lambda/\log\kappa \notin \mathbb{Q}$. □

Applying Lemma 10 with the certified norm values $N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda) = 1/4$ and $N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa) = 1$ established above yields

$$\frac{\log\lambda}{\log\kappa} \notin \mathbb{Q}.$$

8 Exact dominant half-step directions and half-step contractions

The return map $R_G = J_1^G J_0^G$ is the Jacobian of the composition $F_1^G \circ F_0^G$ at the fixed point $\hat{P}_0^G$ (Section 7), and thus governs the first-order dynamics near the 2-cycle, but the proof requires a finer decomposition into half-steps. The phase symmetries from Section 5 relate the two Jacobians at corresponding phases of the 2-cycle. Recall that $F_1^G = S_G \circ F_0^G \circ S_G$ for each gadget. Differentiating at the 2-cycle points yields

$$J_1^A = DS_A\, J_0^A\, DS_A, \qquad J_1^B = DS_B\, J_0^B\, DS_B,$$

where the differentials of the phase-swap maps are the constant matrices

$$DS_A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad DS_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

We first record the abstract mechanism behind the half-step transport: it depends only on the involutive symmetry $F_1 = S \circ F_0 \circ S$ and the simplicity of the dominant eigenvalue.

Lemma 11 (Half-step eigenvector transport)**.** *Let S be an affine involution with constant derivative $C = DS$, $C^2 = I$. Let*

$$F_1 = S \circ F_0 \circ S, \qquad P_1 = S(P_0),$$

and define

$$J_0 = DF_0(P_0), \qquad J_1 = DF_1(P_1).$$

Then

$$J_1 = C J_0 C.$$

Let

$$R = J_1 J_0.$$

Suppose $Rv_0 = \mu v_0$, where $\mu > 0$ is a simple eigenvalue. Define

$$v_1 = C v_0.$$

Then there is a scalar $c \neq 0$ such that

$$J_0 v_0 = c v_1, \qquad J_1 v_1 = c v_0, \qquad c^2 = \mu.$$

If v_0 and v_1 are normalized so their first coordinate is 1, and the first coordinate of $J_0 v_0$ is positive, then

$$c = \sqrt{\mu}.$$

Proof. Differentiating $F_1 = S \circ F_0 \circ S$ at P_1 gives

$$J_1 = CJ_0C.$$

Let

$$R' = J_0J_1.$$

Since $C^2 = I$,

$$CRC = C(J_1J_0)C = (CJ_1C)(CJ_0C) = J_0J_1 = R'.$$

Thus R' is conjugate to R.

Because $Rv_0 = \mu v_0$,

$$R'v_1 = R'(Cv_0) = CRv_0 = \mu Cv_0 = \mu v_1.$$

Also,

$$R'(J_0v_0) = J_0J_1J_0v_0 = J_0Rv_0 = \mu J_0v_0.$$

Since $\mu > 0$, $J_0v_0 \neq 0$. Since μ is simple for R, it is simple for the conjugate matrix R'. Therefore the μ-eigenspace of R' is one-dimensional, and

$$J_0v_0 = cv_1$$

for some $c \neq 0$.

Now

$$J_1v_1 = CJ_0C(Cv_0) = CJ_0v_0 = C(cv_1) = cv_0.$$

Composing,

$$Rv_0 = J_1J_0v_0 = J_1(cv_1) = c^2v_0.$$

Since $Rv_0 = \mu v_0$, $c^2 = \mu$. If the first coordinate of v_1 is 1, then the first coordinate of $J_0v_0 = cv_1$ is c. Positivity of that coordinate gives $c > 0$, hence $c = \sqrt{\mu}$. □

The next lemma is the concrete A/B specialization of Lemma 11, with S specialized to the gadget phase-swap maps S_A, S_B and μ to the dominant eigenvalues λ, κ. The certificate-based positivity input is recorded in the proof.

Lemma 12 (Half-step factorization)**.** *For each gadget $G \in \{A, B\}$, let $R_G = J_1^G J_0^G$ be the full return map with dominant eigenvalue μ_G ($\mu_A = \lambda$, $\mu_B = \kappa$). Let v_0^G be a corresponding eigenvector normalized so that its first entry (the p-component, in the chart coordinates of Section 3) is* 1*, and set $v_1^G = DS_G\, v_0^G$. Then, for $c_G := \sqrt{\mu_G} > 0$, we have*

$$J_0^G v_0^G = c_G v_1^G, \qquad J_1^G v_1^G = c_G v_0^G.$$

Proof. We give the argument for gadget A; gadget B is identical with DS_B, κ replacing DS_A, λ, once we verify that κ is a simple eigenvalue of R_B.

Conjugate return map. Define the *reverse return map* $R'_A = J_0^A J_1^A$. Using $J_1^A = DS_A J_0^A DS_A$ and $DS_A^2 = I$,

$$DS_A R_A DS_A = DS_A J_1^A J_0^A DS_A = (DS_A J_1^A DS_A)(DS_A J_0^A DS_A) = J_0^A J_1^A = R'_A.$$

Hence R'_A is conjugate to R_A and has the same eigenvalues λ, ν. Moreover,

$$R'_A v_1^A = R'_A(DS_A v_0^A) = DS_A R_A v_0^A = \lambda\, DS_A v_0^A = \lambda v_1^A,$$

so v_1^A is the λ-eigenvector of R'_A.

From $R_A v_0^A = \lambda v_0^A$ we get $J_1^A J_0^A v_0^A = \lambda v_0^A$. Therefore, with $w = J_0^A v_0^A$,

$$R'_A w = J_0^A (J_1^A w) = J_0^A (\lambda v_0^A) = \lambda J_0^A v_0^A = \lambda w.$$

Since $\lambda \neq \nu$, the λ-eigenspace of R'_A is one-dimensional. Because both w and v_1^A are λ-eigenvectors, there exists a scalar c_A such that $J_0^A v_0^A = c_A v_1^A$.

The second half-step identity and $c_A^2 = \lambda$. Using $J_1^A = DS_A J_0^A DS_A$, $v_1^A = DS_A v_0^A$, and $DS_A^2 = I$:

$$J_1^A v_1^A = DS_A J_0^A DS_A (DS_A v_0^A) = DS_A J_0^A v_0^A = DS_A (c_A v_1^A) = c_A v_0^A.$$

Composing the two half-steps gives

$$R_A v_0^A = J_1^A (c_A v_1^A) = c_A (c_A v_0^A) = c_A^2 v_0^A = \lambda v_0^A,$$

so, since $v_0^A \neq 0$, $c_A^2 = \lambda$.

Positivity from certified half-step p-multipliers. Because the p-component of v_1^A is exactly 1, the scalar c_A is the p-component of $J_0^A v_0^A$. The certificate (Item 9 of Appendix A.2) computes this p-component directly and proves

$$(J_0^A v_0^A)_p,\ (J_1^A v_1^A)_p \in [0.307395607438829821,\ 0.307395607492351897] \subset (0, 1).$$

Hence $c_A > 0$. Since $c_A^2 = \lambda$, it follows that $c_A = \sqrt{\lambda}$.

Gadget B: simplicity of κ and positivity. As argued above, R_B has distinct eigenvalues λ, ν, κ. Here, κ is a simple eigenvalue of the block-triangular R_B, and the same argument applies. The certificate certifies

$$(J_0^B v_0^B)_p \in [0.451532647302682166,\ 0.451532650145556158],$$

$$(J_1^B v_1^B)_p \in [0.451532647288647226,\ 0.451532650159591153] \subset (0, 1),$$

so $c_B > 0$, and therefore $c_B = \sqrt{\kappa}$. □

A-gadget eigenvectors and contraction

We now characterize eigenvectors for each gadget and verify the half-step contraction. Since $\lambda \neq \nu$, the eigenvalue λ of R_A is simple. Define θ_A as the unique scalar such that $R_A(1, \theta_A)^\top = \lambda(1, \theta_A)^\top$, and set $v_0^A = (1, \theta_A)$. The certificate (Item 8 of Appendix A.2) verifies

$$\theta_A \in [0.580405272518281104,\ 0.580405291461442915].$$

Set $v_1^A = DS_A v_0^A = (1, -\theta_A)$. By Lemma 12, for $c_A := \sqrt{\lambda}$, $J_0^A v_0^A = c_A v_1^A$, $J_1^A v_1^A = c_A v_0^A$. Moreover,

$$c_A \in [0.307395607465570497,\ 0.307395607465570659] < 1.$$

B-gadget eigenvectors and contraction

Since $\kappa \notin \{\lambda, \nu\}$, the eigenvalue κ of R_B is simple. Define $(\theta_{B,0}, \chi_B)$ as the unique pair such that

$$R_B(1, \theta_{B,0}, \chi_B)^\top = \kappa(1, \theta_{B,0}, \chi_B)^\top$$

and set $v_0^B = (1, \theta_{B,0}, \chi_B)$. The certificate (also Item 8 of Appendix A.2) verifies

$$\theta_{B,0} \in [0.198318090249174405,\ 0.198318111627417224], \tag{8.1}$$

$$\chi_B \in [1.01349886578531906,\ 1.01349888527878102]. \tag{8.2}$$

Set

$$v_1^B = DS_B v_0^B = (1, \theta_{B,1}, \chi_B),$$
$$\theta_{B,1} = -(\theta_{B,0} + \chi_B) \in [-1.21181699690619825,\ -1.21181695603449347].$$

By Lemma 12, for $c_B := \sqrt{\kappa}$, $J_0^B v_0^B = c_B v_1^B$, $J_1^B v_1^B = c_B v_0^B$. Moreover,

$$c_B \in [0.451532648724610500,\ 0.451532648724610611] \in (0, 1).$$

9 Explicit phase sectors and their certified invariance

We define explicit cone-like sectors around each phase of the 2-cycle and certify their forward invariance under the branch maps via rigorous interval arithmetic.

Review of interval arithmetic

We briefly discuss the framework. In *interval arithmetic* [6], every real quantity is enclosed by a closed interval $[\underline{x}, \overline{x}] \subset \mathbb{R}$ with $\underline{x} \le \overline{x}$, and each elementary operation $\circ \in \{+, -, \times, \div\}$ is replaced by an interval extension satisfying the *inclusion property*: whenever $x \in [\underline{x}, \overline{x}]$ and $y \in [\underline{y}, \overline{y}]$,

$$x \circ y \ \in\ [\underline{x}, \overline{x}] \circ [\underline{y}, \overline{y}].$$

For example, $[\underline{x}, \overline{x}] + [\underline{y}, \overline{y}] = [\underline{x} + \underline{y}, \overline{x} + \overline{y}]$ and $[\underline{x}, \overline{x}] \times [\underline{y}, \overline{y}] = [\min S, \max S]$ where $S = \{\underline{x}\,\underline{y}, \underline{x}\,\overline{y}, \overline{x}\,\underline{y}, \overline{x}\,\overline{y}\}$; division requires $0 \notin [\underline{y}, \overline{y}]$. By induction on the number of operations, evaluating any rational function f via interval arithmetic on an axis-aligned box $B \subset \mathbb{R}^n$—provided no intermediate divisor interval contains zero—yields an interval I satisfying $f(x) \in I$ for every $x \in B$; that is, $I \supseteq \{f(x) : x \in B\}$.

In our certificate, all interval endpoints are exact rational numbers, so every enclosure is justified. Lemma 8 guarantees pointwise nonvanishing of all denominators arising from the branch maps on the relevant domains. The corresponding interval-arithmetic claim—that no divisor *interval* contains zero in the chosen evaluation scheme on each phase box—does not follow automatically from pointwise nonvanishing, but is verified separately by the certificate. Together, these are the mechanism by which the branch maps, their Jacobians, and their Hessians are enclosed on the phase boxes defined below.

We consider two accuracy parameters r and ρ, where r bounds the principal coordinate (distance from the 2-cycle along the *dominant eigendirection*, which is the direction of v_i^G) and ρ bounds the transverse slope. We set them as $r = 2.5 \times 10^{-8}$, $\rho = 10^{-3}$.

Below we will use standard notation for interval arithmetic, for instance, multiplying intervals with scalars as $a[b, c] = [ab, ac]$, for $a > 0$, $b \le c$. We emphasize that we always use closed intervals $[b, c]$ for interval arithmetic, and thus the notation (b, c) refers to a two-dimensional row vector and not an interval.

A-sectors

For each phase $i \in \{0, 1\}$, the A-sector is the set

$$S_i^A = \Big\{ \hat{P}_i^A + x\, v_i^A + y\,(0, 1) : 0 < x \le r,\ |y| \le \rho x \Big\}.$$

B-sectors

The B-sectors have two transverse directions:

$$S_i^B = \Big\{ \hat{P}_i^B + x\, v_i^B + y\,(0,1,0) + z\,(0,0,1) : 0 < x \le r,\ |y| \le \rho x,\ |z| \le \rho x \Big\}.$$

Recalling the certified interval (8.2), $\chi_B > 1.013$, so $\chi_B - \rho > 1.013 - 10^{-3} > 0$. Hence, every point of S_i^B has $e > 0$, so $S_i^B \subset \Sigma_B^\circ$.

The structural fact behind the adapted-basis first column is abstract:

Lemma 13 (First-column structure in adapted bases)**.** *Under the hypotheses of Lemma 11, let B_i be an adapted basis matrix whose first column is v_i. Then*

$$B_{i\oplus 1}^{-1} J_i B_i$$

has first column

$$(c, 0, \dots, 0)^\top.$$

Proof. The first column of B_i is v_i. By Lemma 11,

$$J_i v_i = c v_{i\oplus 1}.$$

The first column of $B_{i\oplus 1}$ is $v_{i\oplus 1}$, so

$$B_{i\oplus 1}^{-1} J_i v_i = B_{i\oplus 1}^{-1}(c v_{i\oplus 1}) = c e_1.$$

Thus the transformed first column is exactly $(c, 0, \dots, 0)^\top$. □

The concrete A/B form of this structural lemma is the following.

Lemma 14 (Exact first-column structure in the adapted bases)**.** *Define the adapted basis matrices*

$$B_0^A = [v_0^A, (0,1)], \qquad B_1^A = [v_1^A, (0,1)],$$
$$B_0^B = [v_0^B, (0,1,0), (0,0,1)], \qquad B_1^B = [v_1^B, (0,1,0), (0,0,1)].$$

Then, for $i \in \{0,1\}$ (where $i \oplus 1$ denotes addition modulo 2),

$$(B_{i\oplus 1}^A)^{-1} J_i^A B_i^A = \begin{pmatrix} c_A & * \\ 0 & * \end{pmatrix},$$

and

$$(B_{i\oplus 1}^B)^{-1} J_i^B B_i^B = \begin{pmatrix} c_B & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}.$$

Proof. The first column of B_i^A is v_i^A. By Lemma 12, $J_i^A v_i^A = c_A v_{i\oplus 1}^A$. Since $v_{i\oplus 1}^A$ is the first column of $B_{i\oplus 1}^A$, left-multiplying by $(B_{i\oplus 1}^A)^{-1}$ gives first coordinate c_A and zero transverse coordinate. That is exactly the displayed matrix form. The B-case is identical, using $J_i^B v_i^B = c_B v_{i\oplus 1}^B$. □

For each sector S_i^G, a *phase box* is a coordinate-aligned rectangular box (Cartesian product of closed intervals in the chart coordinates) that contains the closure of S_i^G; the certificate uses phase boxes because interval arithmetic operates on such axis-aligned rectangles. The certificate (Item 10

of Appendix A.2) computes interval Jacobians in these exact bases and second-derivative bounds on explicit rational phase boxes covering the sectors.

The B-phase boxes extend to $e = 0$ (their e-coordinate ranges over $[0, k_{Be}r]$); since $Q_B|_{e=0} = Q_A$ and $R_B|_{e=0} = R_A$. Lemma 8 proves *pointwise* regularity: every denominator arising from the branch maps is nonzero at each point of Σ_A° and Σ_B°. Pointwise regularity does not by itself imply that the corresponding *interval enclosure* of a denominator excludes zero on a phase box. The certificate therefore additionally verifies, by exact interval arithmetic on the chosen phase boxes and the explicit evaluation scheme used for each rational function, that every divisor interval encountered along the way excludes zero. This is what makes the interval-arithmetic enclosures of the branch maps, their Jacobians, and their Hessians well-defined on the phase boxes below. By Lemma 14, the lower entries in the first transformed column vanish exactly, so the sector inequalities only use the remaining coefficients and the certified Hessian remainder terms. The general mechanism by which such first-column data plus second-derivative bounds yield sector invariance is the following.

Lemma 15 (Sector invariance from Taylor bounds)**.** *Let $F : U \subset \mathbb{R}^d \to \mathbb{R}^d$ be C^2. Let $P_+ = F(P)$, and let B, B_+ be invertible basis matrices. Define local coordinates*

$$\Phi(u) = B_+^{-1}\big(F(P + Bu) - P_+\big), \qquad u = (x, \eta) \in \mathbb{R} \times \mathbb{R}^{d-1}.$$

Assume

$$P + B\,C(r, \rho) \subset U,$$

where

$$C(r, \rho) = \{(x, \eta) : 0 < x \le r,\ |\eta|_\infty \le \rho x\}.$$

Assume

$$\Phi(0) = 0, \qquad D\Phi(0) = \begin{pmatrix} c & \ell \\ 0 & A \end{pmatrix},$$

with $c > 0$, ℓ a row vector, and A a $(d-1) \times (d-1)$ matrix.

Suppose that for all $u = (x, \eta) \in C(r, \rho)$, the Taylor remainder

$$\Phi(u) = D\Phi(0)u + (R_x(u), R_\eta(u))$$

satisfies

$$|R_x(u)| \le K_x x^2, \qquad |R_\eta(u)|_\infty \le K_\eta x^2.$$

Let

$$a = |\ell|_1, \qquad b = |A|_\infty,$$

where $|A|_\infty$ is the induced row-sum norm. Define

$$m = c - a\rho - K_x r, \qquad M = c + a\rho + K_x r,$$

and

$$\tau_+ = \frac{b\rho + K_\eta r}{m}.$$

If

$$m > 0, \qquad M < 1, \qquad \tau_+ \le \rho,$$

then

$$\Phi(C(r, \rho)) \subset C(r, \rho),$$

and for every $u = (x, \eta) \in C(r, \rho)$, writing $\Phi(u) = (x', \eta')$,

$$m \le \frac{x'}{x} \le M, \qquad \frac{|\eta'|_\infty}{x'} \le \tau_+.$$

Proof. For $u = (x, \eta) \in C(r, \rho)$,

$$x' = cx + \ell\eta + R_x(u).$$

Since $|\eta|_\infty \le \rho x$,

$$|\ell\eta| \le |\ell|_1 |\eta|_\infty \le a\rho x.$$

Also $x \le r$, so

$$|R_x(u)| \le K_x x^2 \le K_x r x.$$

Therefore

$$mx \le x' \le Mx.$$

Since $m > 0$, $x' > 0$. Since $M < 1$ and $x \le r$,

$$x' \le Mx \le Mr < r.$$

For the transverse coordinates,

$$\eta' = A\eta + R_\eta(u),$$

so

$$|\eta'|_\infty \le |A|_\infty |\eta|_\infty + K_\eta x^2 \le (b\rho + K_\eta r)x.$$

Using $x' \ge mx$,

$$\frac{|\eta'|_\infty}{x'} \le \frac{b\rho + K_\eta r}{m} = \tau_+ \le \rho.$$

Thus $\Phi(u) \in C(r, \rho)$. □

The certified half-step bounds listed below are obtained by applying Lemma 15 to each gadget and phase, with the $D\Phi(0)$ first-column structure supplied by Lemma 13 (specialized as Lemma 14) and the Taylor-remainder constants K_x, K_η supplied by the certified Hessian enclosures.

A-sector certified bounds

In the sector coordinates, the ratio x'/x measures the contraction factor along the dominant eigendirection (and should be close to the half-step eigenvalue $c_A \approx 0.3074$), while $|y'|/x'$ measures the transverse drift relative to the cone boundary ρ.

For both phases $i = 0, 1$,

$$\frac{x'}{x} \in [0.307394561088771942,\ 0.307396653842409774],$$
$$\frac{|y'|}{x'} \le 1.10321761561807761 \times 10^{-4} < \rho.$$

(For phase 0 the certified bound is slightly smaller, $1.10321759830186699 \times 10^{-4}$.)

B-sector certified bounds

At phase 0:

$$\frac{x'}{x} \in [0.451388989838190632,\ 0.451676307610047724],$$
$$\frac{\max(|y'|, |z'|)}{x'} \le 6.82589234315899433 \times 10^{-4} < \rho.$$

At phase 1:

$$\frac{x'}{x} \in [0.451389955166172293,\ 0.451675342282066063],$$
$$\frac{\max(|y'|,|z'|)}{x'} \le 6.84951234683088803 \times 10^{-4} < \rho.$$

The A-gadget bounds above are uniform over both phases (the phase-0 bound on $|y'|/x'$ is slightly tighter, as noted in the parenthetical), so they are reported together. The B-gadget's 3-dimensional dynamics make the two phases slightly asymmetric, so we report them separately above.

Therefore

$$F_i^A(S_i^A) \subset S_{i\oplus 1}^A, \qquad F_i^B(S_i^B) \subset S_{i\oplus 1}^B.$$

Define the transverse ratios $\tau_m^A = |y_m|/x_m^A$ and $\tau_m^B = \max(|y_m|,|z_m|)/x_m^B$. The same computation yields explicit transverse-ratio recurrences, which will be needed for the asymptotic analysis in Section 12:

$$\begin{aligned}
&\tau_{m+1}^A \le \alpha_A \tau_m^A + \beta_A x_m^A, \\
&\qquad \alpha_A \le 0.110098192661922619, \quad \beta_A \le 8.94275599540566418, \\
&\tau_{m+1}^B \le \alpha_B \tau_m^B + \beta_B x_m^B, \\
&\qquad \alpha_B \le 0.684664374851195431, \quad \beta_B \le 12.8511069222196774,
\end{aligned}$$

Here m indexes the successive bursts won by each respective gadget. Since $\alpha_A, \alpha_B < 1$ and the principal coordinates are geometrically summable, the transverse-ratio series $\sum_m \tau_m^A$ and $\sum_m \tau_m^B$ converge; the full argument is given in Section 12.

10 Local branch words and whole-burst locking

Within the invariant sectors, we must verify two properties: that the column-selection sequence within each gadget matches the expected half-word (H_0 or H_1), and that once a gadget wins the first round of a 5-step burst, it wins all five rounds. Both are established by certified gap bounds.

The certificate (Items 11–12 of Appendix A.2) checks that at each step of each half-word on the full phase boxes, the corresponding column has the largest edge among the four columns of the active gadget. The minimum margin is

$$g_{\mathrm{br}}^{\mathrm{box}} > 0.0250983508506417014.$$

It also certifies that at steps $2,3,4,5$ of each half-word the selected edge exceeds $\mu_* = 2p_* - 1$ by at least

$$\Delta_{\mathrm{box}} > 0.0758691457701749199.$$

The largest possible first-edge surplus on any phase box is only

$$\delta_{\mathrm{first}}^{\max} < 1.05687287632489371 \times 10^{-7}.$$

Consequently, at steps $2,3,4,5$ of any burst the active gadget's selected edge exceeds the best edge of the frozen gadget by at least $\Delta_{\mathrm{box}} - \delta_{\mathrm{first}}^{\max}$ (which we denote as the *whole-burst lock margin*):

$$\Delta_{\mathrm{box}} - \delta_{\mathrm{first}}^{\max} > 0.0758690400828872874.$$

We now formalize the locking implication abstractly.

Lemma 16 (Whole-burst locking)**.** *Assume the following finite gap assertions hold on every relevant phase box.*

1. *At every step of the prescribed half-word H_i, the prescribed active-gadget column exceeds every other column in the active gadget by at least $g > 0$.*
2. *At steps $2, 3, 4, 5$, the prescribed active-gadget edge is at least $\mu_* + \Delta$.*
3. *At a burst boundary, every column edge of the frozen gadget is at most $\mu_* + \delta$.*
4. *$\Delta > \delta$.*

Then once a gadget wins the first step of a burst, it wins all five steps. During that burst, the active gadget evolves by the corresponding branch map F_i^G, and the inactive gadget is unchanged.

Proof. Assume A wins the first step. By condition 1, the first prescribed A-column is the unique best A-column. Since it wins globally at the first step, the A-state is updated by the first letter of the prescribed word.

Inductively suppose the first $k-1$ steps, with $2 \le k \le 5$, have followed the prescribed A-word and selected only A-columns. Then the B-state has not moved, so all B-edges are still their burst-boundary values. By condition 3, every B-edge is at most $\mu_* + \delta$. By condition 2, the prescribed A-edge at step k is at least $\mu_* + \Delta$, which is strictly larger because $\Delta > \delta$. By condition 1, no other A-column beats the prescribed A-column. Hence the prescribed A-column wins step k globally.

This proves that A wins all five steps and realizes the branch map F_i^A. The proof for B is identical. □

The certified numerical inequalities $g_{\mathrm{br}}^{\mathrm{box}} > 0$, $\Delta_{\mathrm{box}} > \delta_{\mathrm{first}}^{\max}$ above instantiate the hypotheses of Lemma 16, and the conclusion follows. As an immediate consequence, the column-selection sequence within the A-gadget is exactly H_0 on S_0^A and exactly H_1 on S_1^A; likewise for the B-gadget on S_0^B and S_1^B.

11 Product burst dynamics

Let $i_n \in \{0,1\}$ and $j_n \in \{0,1\}$ denote the A-phase and B-phase at burst boundary n, respectively. Because the starting point lies in $S_0^A \times S_0^B$ (Section 14), the certified sector invariance $F_i^G(S_i^G) \subset S_{i\oplus 1}^G$ from Section 9 and the whole-burst lock from Section 10 imply by induction that $\hat{a}_n \in S_{i_n}^A$ and $\hat{b}_n \in S_{j_n}^B$ for every burst boundary n.

Let $W_n = a_n \otimes b_n \in \mathcal{P}$ be the burst-boundary states, where $a_n \in \Delta^3$ and $b_n \in \Delta^4$ are the simplex marginals. Write the corresponding chart coordinates as

$$\begin{aligned} \hat{a}_n &= \hat{P}_{i_n}^A + x_n^A v_{i_n}^A + y_n^A(0,1), \\ \hat{b}_n &= \hat{P}_{j_n}^B + x_n^B v_{j_n}^B + y_n^B(0,1,0) + z_n^B(0,0,1), \end{aligned} \tag{11.1}$$

with $\hat{a}_n \in S_{i_n}^A$ and $\hat{b}_n \in S_{j_n}^B$. For brevity write $x_n := x_n^A$, $s_n := x_n^B$.

Lemma 17 (First-edge comparison at burst boundaries)**.** *In either gadget and either phase, the first column of the half-word has edge*

$$2p - 1 = \mu_* + 2(p - p_*).$$

In the adapted sector coordinates, $p - p_* = x$. Therefore the first prescribed edge in the A-gadget is $\mu_* + 2x_A$, and the first prescribed edge in the B-gadget is $\mu_* + 2x_B$.

If the certified local branch gap says this first prescribed column is the unique local maximizer in each gadget sector, then the first global step chooses A iff $x_A \geq x_B$, with equality resolved in favor of A because the A-columns precede the B-columns.

Proof. For L_A, in the chart

$$a(p, d) = \left(p - \tfrac{1}{2},\ 1 - p,\ \tfrac{1}{2} - d,\ d\right),$$

the first column is $(1, -1, 1, 1)^\top$, so its edge is

$$\left(p - \tfrac{1}{2}\right) - (1 - p) + \left(\tfrac{1}{2} - d\right) + d = 2p - 1.$$

For L_B,

$$b(p, d, e) = \left(p - \tfrac{1}{2},\ 1 - p,\ \tfrac{1}{2} - d - e,\ d,\ e\right),$$

and the first column is $(1, -1, 1, 1, 1)^\top$, so its edge is

$$\left(p - \tfrac{1}{2}\right) - (1 - p) + \left(\tfrac{1}{2} - d - e\right) + d + e = 2p - 1.$$

The adapted dominant vector is normalized to have p-coordinate 1, and the transverse basis vectors have p-coordinate 0. Hence $p - p_* = x$ in sector coordinates.

Thus the best local first edge is $\mu_* + 2x_A$ for A and $\mu_* + 2x_B$ for B, assuming the local first-column branch gap has certified that these are indeed the unique local maximizers. The global comparison is therefore exactly the comparison between x_A and x_B. If equality holds, smallest-index tie-breaking selects the A-column because the A-columns are listed before the B-columns. □

By Lemma 17, the first global round chooses

- A if $x_n > s_n$;
- B if $s_n > x_n$;
- A on equality, by smallest-index tie-breaking, because the A-columns have smaller global indices than the B-columns.

Because of the whole-burst lock margin from Section 10, that first-round winner then wins the entire 5-step burst.

Half-step contraction bounds

The certified sector bounds from Section 9 yield the following contraction bounds. If A wins burst n, then $x_{n+1} \in [m_A, M_A]\, x_n, s_{n+1} = s_n$, with

$$m_A = 0.307394561088771942, \qquad M_A = 0.307396653842409774.$$

If B wins burst n, then $s_{n+1} \in [m_B, M_B]\, s_n, x_{n+1} = x_n$, with

$$m_B = 0.451388989838190632, \qquad M_B = 0.451676307610047724.$$

Writing $r_n = x_n/s_n$, the contraction bounds above translate to: if A wins burst n, then $r_{n+1} \in [m_A r_n,\ M_A r_n]$; if B wins, then $r_{n+1} \in [r_n/M_B,\ r_n/m_B]$.

Since A wins precisely when $r_n \geq 1$ and B wins when $r_n < 1$, the ratio r_n has a multiplicative dynamics that we can summarize as follows. We first state an abstract lemma in terms of arbitrary contraction intervals; the concrete case follows by specializing to the certified intervals $[m_A, M_A]$ and $[m_B, M_B]$.

Lemma 18 (Run-length bounds from contraction intervals)**.** *Let $x_n, s_n > 0$ be the A- and B-principal coordinates at burst boundary n, and set*

$$r_n = \frac{x_n}{s_n}.$$

Assume A wins iff $r_n \geq 1$, with equality going to A, and B wins iff $r_n < 1$. Suppose

$$\begin{aligned} &A\text{-win:} \qquad x_{n+1} \in [m_A, M_A]x_n, \quad s_{n+1} = s_n, \\ &B\text{-win:} \qquad s_{n+1} \in [m_B, M_B]s_n, \quad x_{n+1} = x_n, \end{aligned}$$

where

$$0 < m_A \leq M_A < 1, \qquad 0 < m_B \leq M_B < 1.$$

Assume

$$\frac{M_A}{m_B} < 1, \qquad \frac{m_A}{M_B^2} > 1.$$

Then both burst types occur infinitely often. After a finite initial prefix,

$$r_n \in [m_A, 1/m_B],$$

every A-run has length exactly 1, every B-run has length at most 2, and

$$x_n + s_n \to 0.$$

Proof. If only A-bursts occurred from some point on, then

$$r_n \leq M_A^{n-n_0} r_{n_0} \to 0,$$

eventually forcing $r_n < 1$, hence a B-burst. Contradiction. If only B-bursts occurred from some point on, then

$$r_n \geq \frac{r_{n_0}}{M_B^{n-n_0}} \to \infty,$$

eventually forcing $r_n \geq 1$, hence an A-burst. Thus both burst types occur infinitely often.

After a B-burst with $r_n < 1$,

$$r_{n+1} \leq \frac{r_n}{m_B} < \frac{1}{m_B}.$$

After an A-burst with $r_n \geq 1$,

$$r_{n+1} \geq m_A r_n \geq m_A.$$

Once both types have occurred, the ratio enters and remains in $[m_A, 1/m_B]$.

Now suppose A wins at step n, so $r_n \geq 1$. In the invariant interval, $r_n \leq 1/m_B$. Therefore

$$r_{n+1} \leq M_A r_n \leq \frac{M_A}{m_B} < 1.$$

So the next burst is B. Hence every A-run has length exactly 1.

Suppose B wins at step n, so $r_n < 1$. In the invariant interval, $r_n \geq m_A$. After one B-burst,

$$r_{n+1} \geq \frac{r_n}{M_B} \geq \frac{m_A}{M_B}.$$

If $r_{n+1} \geq 1$, the B-run ends after one burst. If $r_{n+1} < 1$, then B wins again, and

$$r_{n+2} \geq \frac{r_{n+1}}{M_B} \geq \frac{m_A}{M_B^2} > 1.$$

Thus the B-run ends after at most two bursts.

Therefore every three consecutive bursts contain at least one A-burst and at least one B-burst. In any such block,

$$x_{n+3} \leq M_A x_n, \qquad s_{n+3} \leq M_B s_n.$$

Hence, with $\theta = \max(M_A, M_B) < 1$,

$$x_{n+3} + s_{n+3} \leq \theta(x_n + s_n).$$

This implies $x_n + s_n \to 0$. □

Lemma 19 (Burst-ratio dynamics and run-length bounds)**.** *For all sufficiently large n, one has $r_n \in [m_A,\, 1/m_B]$. Moreover:*

(i) every A-run has length exactly 1;

(ii) every B-run has length at most 2.

Consequently, every block of three consecutive bursts contains at least one A-burst and at least one B-burst, and $x_n + s_n \to 0$.

Proof. This is the specialization of Lemma 18 to the certified intervals $[m_A, M_A]$ and $[m_B, M_B]$ above; the certified bounds $M_A/m_B < 1$ and $m_A/M_B^2 > 1$ are verified numerically. We retain the direct proof for transparency.

Entry into the compact interval. After a B-burst at step n (so $r_n < 1$), we have $r_{n+1} \leq r_n/m_B < 1/m_B$. After an A-burst at step n (so $r_n \geq 1$), we have $r_{n+1} \geq m_A r_n \geq m_A$. Once the trajectory has experienced at least one A-burst and one B-burst, r_n enters and remains in $[m_A, 1/m_B]$. Both burst types must eventually occur: if only A-bursts occurred, then $r_n \leq M_A^n r_0 \to 0$ (since $M_A < 1$), eventually forcing $r_n < 1$ and hence a B-burst; symmetrically, if only B-bursts occurred, then $r_n \geq r_0/M_B^n \to \infty$ (since $M_B < 1$), eventually forcing $r_n \geq 1$ and hence an A-burst.

A-runs have length 1. Suppose A wins at step n, so $r_n \geq 1$, and that $r_n \leq 1/m_B$ (since we are in the compact interval). Then,

$$r_{n+1} \leq M_A r_n \leq \frac{M_A}{m_B} < 1,$$

where the last inequality is the certified bound $M_A/m_B < 1$. Hence B wins at step $n+1$, and every A-run has length exactly 1.

B-runs have length at most 2. Suppose B wins at step n, so $r_n < 1$, and that $r_n \geq m_A$. After one B-burst,

$$r_{n+1} \geq \frac{r_n}{M_B} \geq \frac{m_A}{M_B}.$$

If $r_{n+1} \geq 1$, the B-run ends after one step. If $r_{n+1} < 1$, then B wins again, and

$$r_{n+2} \geq \frac{r_{n+1}}{M_B} \geq \frac{m_A}{M_B^2} > 1,$$

where the last inequality is the certified bound $m_A/M_B^2 > 1$. Hence A wins at step $n+2$, and every B-run has length at most 2.

Exponential convergence. By Steps 2 and 3, every block of three consecutive bursts contains at least one A-burst and at least one B-burst, so

$$x_{n+3} + s_{n+3} \le \theta\,(x_n + s_n), \qquad \theta = \max(M_A, M_B) < 1.$$

This finishes the proof. □

12 Local asymptotics on the two gadgets

We now establish that the principal coordinates along each gadget's dominant eigendirection contract at rates asymptotically governed by $c_A = \sqrt{\lambda}$ and $c_B = \sqrt{\kappa}$, with summable relative errors. Let $m \mapsto \hat{a}^{(m)}$ be the chart coordinates of the local A-state immediately before the mth A-burst, and $m \mapsto \hat{b}^{(m)}$ the chart coordinates of the local B-state immediately before the mth B-burst. Write

$$x_m^A = p(\hat{a}^{(m)}) - p_*, \qquad x_m^B = p(\hat{b}^{(m)}) - p_*.$$

Because the exact basis vectors from Section 8 satisfy $J_i^A v_i^A = c_A v_{i\oplus 1}^A$, $J_i^B v_i^B = c_B v_{i\oplus 1}^B$, the certified Taylor bounds from Section 9 give

$$\begin{aligned} x_{m+1}^A &= c_A x_m^A\big(1+\varepsilon_m^A\big), & |\varepsilon_m^A| &\le \Gamma_A \tau_m^A + \Delta_A x_m^A, \\ x_{m+1}^B &= c_B x_m^B\big(1+\varepsilon_m^B\big), & |\varepsilon_m^B| &\le \Gamma_B \tau_m^B + \Delta_B x_m^B, \end{aligned}$$

with certified constants

$$\begin{aligned} \Gamma_A &\le 0.003273115711042833, & \Delta_A &\le 5.232171784223787, \\ \Gamma_B &\le 0.31803397447249865, & \Delta_B &\le 8.463704731868965, \end{aligned}$$

and therefore uniform relative-error bounds

$$\begin{aligned} \Gamma_A \tau + \Delta_A x &\le 0.0000034039200056484279 1 && \text{throughout the A sectors,} \\ \Gamma_B \tau + \Delta_B x &\le 0.00031824556709079535 0 && \text{throughout the B sectors.} \end{aligned}$$

In particular, the relative errors are everywhere < 0.01.

Corollary 20 (Summable transverse and relative errors)**.** *Suppose a local sequence in sectors satisfies*

$$\begin{aligned} x_{m+1} &= c\,x_m(1+\varepsilon_m), \\ |\varepsilon_m| &\le \Gamma\tau_m + \Delta x_m, \\ \tau_{m+1} &\le \alpha\tau_m + \beta x_m, \end{aligned}$$

with

$$0 < c < 1, \qquad 0 \le \alpha < 1, \qquad x_{m+1} \le M x_m, \qquad 0 < M < 1.$$

Assume also $|\varepsilon_m| \le \eta < 1/2$. *Then*

$$\sum_m x_m < \infty, \qquad \sum_m \tau_m < \infty, \qquad \sum_m |\varepsilon_m| < \infty,$$

and

$$\log x_m = \log x_0 + m \log c + O(1).$$

Proof. The bound $x_{m+1} \le M x_m$ gives $x_m \le M^m x_0$, so $\sum_m x_m < \infty$. Iterating the transverse recurrence gives

$$\tau_m \le \alpha^m \tau_0 + \beta \sum_{k=0}^{m-1} \alpha^{m-1-k} x_k.$$

Summing over m and exchanging sums,

$$\sum_m \tau_m \le \frac{\tau_0}{1-\alpha} + \frac{\beta}{1-\alpha} \sum_k x_k < \infty.$$

Therefore

$$\sum_m |\varepsilon_m| \le \Gamma \sum_m \tau_m + \Delta \sum_m x_m < \infty.$$

Since $|\varepsilon_m| \le \eta < 1/2$, there is a constant C_η such that $|\log(1+\varepsilon_m)| \le C_\eta |\varepsilon_m|$. Thus $\sum_m \log(1+\varepsilon_m)$ converges absolutely. Hence

$$\log x_m = \log x_0 + m \log c + \sum_{k=0}^{m-1} \log(1+\varepsilon_k) = \log x_0 + m \log c + O(1).$$

□

Applying Corollary 20 to each of the two local sequences (with the certified $M_A, M_B < 1$, the transverse contraction rates $\alpha < 1$ from Section 9, and $\eta = 0.01$), we conclude

$$\sum_m |\varepsilon_m^A| < \infty, \qquad \sum_m |\varepsilon_m^B| < \infty,$$

and

$$\log x_m^A = \log x_0^A + m \log c_A + O(1),$$
$$\log x_m^B = \log x_0^B + m \log c_B + O(1).$$

Let $N_A(n)$ denote the number of A-bursts among bursts $0, \dots, n-1$, and let $N_B(n) = n - N_A(n)$. At burst boundary n, the A-state has been updated exactly $N_A(n)$ times and the B-state exactly $N_B(n)$ times, so

$$\log x_n = \log x_0 + N_A(n) \log c_A + O(1),$$
$$\log s_n = \log s_0 + N_B(n) \log c_B + O(1).$$

Because $r_n = x_n/s_n$ stays in the compact interval $[m_A, 1/m_B]$, the logarithmic difference is bounded:

$$N_A(n) \log c_A - N_B(n) \log c_B = O(1).$$

Since $N_A(n) + N_B(n) = n$, it follows that

$$\frac{N_A(n)}{n} \to \frac{-\log c_B}{-\log c_A - \log c_B} = \frac{-\log \kappa}{-\log \lambda - \log \kappa}.$$

We now isolate this conclusion as a formal proposition.

Proposition 21 (Irrational burst frequency)**.** *Let $N_A(n)$ and $N_B(n) = n - N_A(n)$ be the numbers of A- and B-bursts among the first n bursts. Suppose*

$$\log x_n = \log x_0 + N_A(n) \log c_A + O(1),$$

$$\log s_n = \log s_0 + N_B(n) \log c_B + O(1),$$

where $0 < c_A, c_B < 1$, and suppose x_n/s_n stays bounded above and below away from 0. Then

$$\frac{N_A(n)}{n} \to \frac{-\log c_B}{-\log c_A - \log c_B}.$$

If

$$c_A = \sqrt{\lambda}, \qquad c_B = \sqrt{\kappa},$$

and $\log \lambda / \log \kappa \notin \mathbb{Q}$, then the limiting frequency is irrational. Consequently, the burst-winner sequence is not eventually periodic.

Proof. Since x_n/s_n is bounded above and below,

$$\log x_n - \log s_n = O(1).$$

Subtracting the two logarithmic asymptotics gives

$$N_A(n) \log c_A - N_B(n) \log c_B = O(1).$$

Let $a = -\log c_A > 0$ and $b = -\log c_B > 0$. Because $N_B(n) = n - N_A(n)$,

$$-aN_A(n) + b(n - N_A(n)) = O(1).$$

Thus

$$(a + b)N_A(n) = bn + O(1),$$

and

$$\frac{N_A(n)}{n} \to \frac{b}{a+b} = \frac{-\log c_B}{-\log c_A - \log c_B}.$$

If $c_A = \sqrt{\lambda}$ and $c_B = \sqrt{\kappa}$, then

$$\frac{-\log c_B}{-\log c_A - \log c_B} = \frac{-\log \kappa}{-\log \lambda - \log \kappa}.$$

If this number were rational, then $-\log \kappa/(-\log \lambda - \log \kappa) = q \in \mathbb{Q}$. Since $0 < q < 1$,

$$(1 - q)(-\log \kappa) = q(-\log \lambda),$$

so

$$\frac{\log \lambda}{\log \kappa} = \frac{1-q}{q} \in \mathbb{Q},$$

contradicting Lemma 10. Therefore the limiting A-burst frequency is irrational.

Every eventually periodic binary sequence has a rational limiting symbol frequency, equal to the number of A's in one eventual period divided by the period length. Hence the burst-winner sequence is not eventually periodic. □

13 No finite-cycle convergence

We now combine the preceding results to show that the AdaBoost orbit on M_0 does not converge to any finite cycle. The argument proceeds by contradiction: assuming cycle convergence, we deduce that the burst-winner sequence is eventually periodic, contradicting its irrational asymptotic frequency.

Recalling (6.1), (6.2), define $Q_{ij} = P_i^A \otimes P_j^B$, $i, j \in \{0, 1\}$. These four points are distinct: the tensor product $P_i^A \otimes P_j^B$ determines its marginals by summation, and $P_0^A \neq P_1^A$, $P_0^B \neq P_1^B$ (their 3rd and 4th coordinates are swapped, with $d_* \neq \frac{1}{4}$). Since $x_n, s_n \to 0$, the burst-boundary states satisfy $\|W_n - Q_{i_n j_n}\| \to 0$. Indeed, first recall that $W_n = a_n \otimes b_n \in \mathcal{P}$. Then, mapping the sector parametrization (11.1) to the affine chart gives $\|a_n - P_{i_n}^A\|_1 = O(x_n)$ and $\|b_n - P_{j_n}^B\|_1 = O(s_n)$, while for distributions $\|a \otimes b - a' \otimes b'\|_1 \leq \|a - a'\|_1 + \|b - b'\|_1$; the claim then follows from $x_n + s_n \to 0$ (Lemma 19).

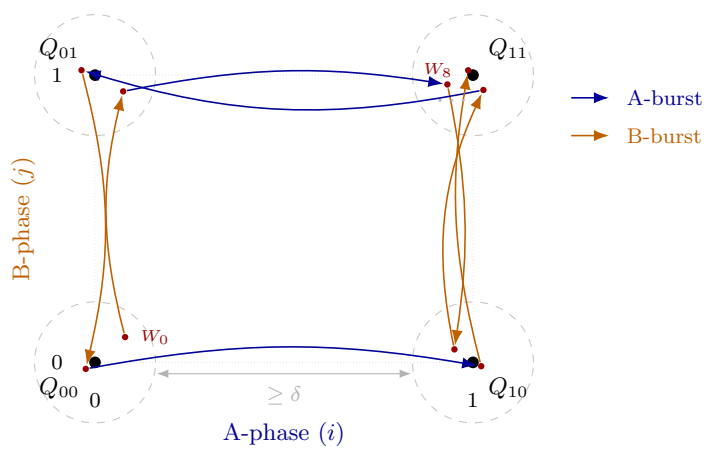
Proposition 22 (Aperiodic burst sequence rules out finite-cycle convergence)**.** *Let W_n be the burst-boundary states, and suppose there are four distinct points*

$$Q_{ij}, \qquad i, j \in \{0, 1\},$$

such that

$$\|W_n - Q_{i_n j_n}\| \to 0,$$

where (i_n, j_n) is the phase-label sequence. Suppose also that each burst toggles exactly one phase coordinate: an A-burst toggles i_n, and a B-burst toggles j_n. If the full AdaBoost trajectory converged to a finite cycle, then the burst-winner sequence would be eventually periodic.



$$\frac{N_A(n)}{n} \to \frac{-\log \kappa}{-\log \lambda - \log \kappa} \notin \mathbb{Q} \implies \text{burst-winner sequence is aperiodic}$$

Figure 2: Schematic of the no-finite-cycle argument. The four points $Q_{ij} = P_i^A \otimes P_j^B$ sit at the vertices of a phase square in $\mathcal{P}$, pairwise separated by at least $\delta > 0$. Each A-burst (blue) toggles the A-phase i, and each B-burst (orange) toggles the B-phase j. The burst-boundary states W_n (red dots) converge toward these vertices as $\|W_n - Q_{i_n j_n}\| \to 0$, but the irrational limiting frequency of A-bursts precludes eventual periodicity of the burst-winner sequence and hence convergence of the AdaBoost orbit to any finite cycle.

Proof. Assume the full AdaBoost trajectory converges to a cycle of length ℓ. Since each burst consists of five AdaBoost steps, the burst-boundary subsequence W_n converges along residue classes modulo

$$L = \frac{\ell}{\gcd(\ell, 5)}.$$

That is, for each $r \in \{0, \dots, L-1\}$,

$$W_{kL+r} \to \widetilde{W}_r$$

for some limit $\widetilde{W}_r$.

Since the four Q_{ij} are distinct, define

$$\delta = \min\{\|Q - Q'\| : Q, Q' \in \{Q_{00}, Q_{01}, Q_{10}, Q_{11}\},\ Q \neq Q'\} > 0.$$

Fix a residue r, and write

$$Q_k = Q_{i_{kL+r}\, j_{kL+r}}.$$

Because

$$W_{kL+r} \to \widetilde{W}_r$$

and

$$\|W_{kL+r} - Q_k\| \to 0,$$

there is K_r such that for all $k \geq K_r$,

$$\|W_{kL+r} - \widetilde{W}_r\| < \delta/4, \qquad \|W_{kL+r} - Q_k\| < \delta/4.$$

For $k, k' \geq K_r$,

$$\begin{aligned} \|Q_k - Q_{k'}\| &\leq \|Q_k - W_{kL+r}\| + \|W_{kL+r} - \widetilde{W}_r\| \\ &\quad + \|\widetilde{W}_r - W_{k'L+r}\| + \|W_{k'L+r} - Q_{k'}\| \\ &< \delta. \end{aligned}$$

By definition of δ, distinct Q-points are at least δ apart, so $Q_k = Q_{k'}$. Thus (i_{kL+r}, j_{kL+r}) is eventually constant on each residue class r.

Therefore (i_n, j_n) is eventually periodic. Since each burst toggles exactly one coordinate, the winner at burst n is determined by the transition

$$(i_n, j_n) \to (i_{n+1}, j_{n+1}).$$

Thus the burst-winner sequence is also eventually periodic. □

By Proposition 22, convergence of the full AdaBoost trajectory to any finite cycle would force the burst-winner sequence (i_n, j_n) to be eventually periodic. This contradicts Section 12 (Proposition 21), which showed that the burst-winner sequence has an irrational limiting frequency. Therefore the AdaBoost orbit on M_0 does not converge to any finite cycle.

14 Explicit rational start and explicit uniform-start counterexample

It remains to exhibit a concrete starting point inside the invariant sectors and to convert it into a uniform-start counterexample. We begin by taking

$$A = (137116847818847,\ 362883152181153,\ 242004611239991,\ 257995388760009),$$
$$B = (137116857552466,\ 362883142447534,\ 242004593130510,\ 257995386869490,\ 20000000).$$

Since simple arithmetic shows that $\sum_i A_i = \sum_j B_j = 10^{15}$, set $a_0 = \frac{A}{10^{15}}$, $b_0 = \frac{B}{10^{15}}$, $W_0 = a_0 \otimes b_0$. To apply the sector-based analysis of Sections 9–13, we need to verify that this starting point lies inside the invariant sectors.

The certificate (Item 13 of Appendix A.2) proves that the chart images satisfy $\hat{a}_0 \in S_0^A$, $\hat{b}_0 \in S_0^B$. The starting point is decomposed in sector coordinates: we report the principal coordinate x (distance from the 2-cycle along the dominant eigendirection) and the transverse ratios (deviation from the eigendirection, normalized by x), confirming that they satisfy the sector constraint $|\text{transverse ratio}| \leq \rho$.

More explicitly: $x_A \in [9.999999 \times 10^{-9},\ 1.0000001 \times 10^{-8}]$, with A-sector residual in

$$[-2.18994134061382760 \times 10^{-4},\ 2.19798676726244218 \times 10^{-4}],$$

and $x_B \in [1.9733618 \times 10^{-8},\ 1.9733620 \times 10^{-8}]$, with B d-residual in

$$[-1.10962941057972692 \times 10^{-4},\ 1.11368115987132873 \times 10^{-4}],$$

and B e-residual in

$$[-9.48895873525338312 \times 10^{-8},\ 2.73218648306331209 \times 10^{-8}],$$

all strictly inside the sector slope bound $\rho = 10^{-3}$. Note that $x_A \approx 10^{-8} < 1.97 \times 10^{-8} \approx x_B$, so $r_0 = x_A/x_B < 1$ and the first burst is a B-burst.

Lemma 23 (Row duplication and class aggregation)**.** *Let M' be obtained from M by duplicating rows. Partition the rows of M' into duplicate classes, one class for each row of M. Let D'_t be an AdaBoost trajectory on M', and let $\bar{D}_t$ be the class-summed distribution on M. Then:*

1. *The edge vector of D'_t on M' equals the edge vector of $\bar{D}_t$ on M.*
2. *The class-summed weights $\bar{D}_t$ evolve exactly as AdaBoost on M.*
3. *If D'_0 is uniform on M', then $\bar{D}_0$ assigns to each original row a mass proportional to its number of duplicates.*
4. *$\gamma(M') = \gamma(M)$.*

Proof. Rows in a duplicate class are identical. Therefore, for any column, the contribution of a class to the edge depends only on the sum of the weights in that class. This proves the edge-vector identity. Since the selected edge vector is the same, the selected column is the same under the same tie-breaking rule.

For a selected column j, every row in a duplicate class has the same sign M_{ij}, so every row in that class is multiplied by the same update factor

$$\frac{1}{1 + r_t M_{ij}}.$$

Therefore the class sum is updated by exactly the same formula as the corresponding row weight in M. This proves the dynamical aggregation claim.

If D'_0 is uniform, the class sum for an original row is exactly

$$\frac{\text{number of duplicates of that row}}{\text{total number of duplicated rows}}.$$

For the margin, every distribution on M' aggregates to a distribution on M with the same edge vector, and every distribution on M lifts to a distribution on M' with the same edge vector. Thus the set of attainable edge vectors is the same, and

$$\gamma(M') = \gamma(M).$$

□

Lemma 24 (Augmented negated-column margin)**.** *Let*

$$M^{\pm} = [M \mid -M].$$

Suppose $\gamma(M) = \gamma > 0$, and suppose there exists a distribution D_* such that every positive-column edge satisfies

$$(D_*^\top M)_j = \gamma.$$

Then

$$\gamma(M^{\pm}) = \gamma.$$

Proof. For every distribution D,

$$\max_{h \in M^{\pm}} D^\top h \geq \max_{h \in M} D^\top h \geq \gamma(M) = \gamma.$$

Thus

$$\gamma(M^{\pm}) \geq \gamma.$$

At D_*, every positive-column edge is γ, and every negated-column edge is $-\gamma$. Hence

$$\max_{h \in M^{\pm}} D_*^\top h = \gamma,$$

so

$$\gamma(M^{\pm}) \leq \gamma.$$

Therefore

$$\gamma(M^{\pm}) = \gamma.$$

□

Lemma 25 (No-negated-column reduction)**.** *Let $M^{\pm} = [M \mid -M]$, with the original columns ordered first and the negated columns appended afterward. Suppose a trajectory on M has selected positive-column edge r_t at time t, and suppose*

$$r_t > -\min_j (D_t^\top M)_j$$

for every t. Then the trajectory on $M^{\pm}$, started from the same distribution and using the same tie-breaking order on the original columns, is identical to the trajectory on M.

Proof. The edge of the negated column $-M_{\cdot j}$ is

$$D_t^\top(-M_{\cdot j}) = -(D_t^\top M_{\cdot j}).$$

Thus the best negated-column edge is

$$\max_j D_t^\top(-M_{\cdot j}) = -\min_j (D_t^\top M_{\cdot j}).$$

By hypothesis, this is strictly smaller than the selected positive-column edge r_t. Therefore no negated column belongs to the argmax set. The selected column in $M^{\pm}$ is the same positive column selected in M. The update formula is then identical.

Induction over t proves the trajectories agree at all times. □

Now construct $\widetilde{M}$ by replacing each row (i, j) of M_0 by $A_i B_j$ identical copies. Since $\sum_i A_i \sum_j B_j = 10^{30}$, the exact uniform distribution on $\widetilde{M}$ aggregates to W_0. By Lemma 23, the class-aggregated weights of the uniform-start AdaBoost orbit on $\widetilde{M}$ correspond exactly to the weights on the W_0-started orbit on M_0. Since the latter does not converge to a finite cycle, neither does the former.

By Lemma 23 and Section 4, $\gamma(\widetilde{M}) = \gamma(M_0) = \frac{1}{5}$. This completes the proof of Theorem 1. □

Proof of Corollary 2. Set $\widetilde{M}' = [\widetilde{M} \mid -\widetilde{M}]$, where the columns of $-\widetilde{M}$ are appended after those of $\widetilde{M}$. By construction, the column set of $\widetilde{M}'$ is closed under negation.

We verify the strict-edge hypothesis of Lemma 25 on the trajectory on $\widetilde{M}$: at every step the selected positive-column edge strictly exceeds $-\min_j (D_t^\top \widetilde{M})_j$, equivalently strictly exceeds every negated-column edge. By Lemma 23, the edge vectors on $\widetilde{M}$ equal those on M_0 at every step, so it suffices to verify on M_0 that the selected column edge strictly exceeds every negated column edge at every step. The certificate does so by exact interval arithmetic on the full sector boxes (items 15–16 of Appendix A.2). During any burst, two types of negated columns could compete: those from the active gadget and those from the frozen (non-active) gadget.

For the frozen gadget, Lemma 3 implies its column edges do not change during the burst; the certificate encloses all frozen-gadget column edges on each sector box. For the active gadget, at each of the five steps within each burst word on each sector box, the certificate (Item 15 of Appendix A.2) encloses all column edges. At every step of every burst/frozen-phase combination, the certificate (Item 15 of Appendix A.2) verifies that the selected edge strictly exceeds every negated column edge. The minimum certified gap over all eight burst/frozen-phase combinations and all five steps is

$$g_{\text{neg}}^{\text{box}} \geq 0.274233539263122735 > 0,$$

which certifies the strict inequality required by Lemma 25. Hence by Lemma 25 the trajectory on $\widetilde{M}'$ coincides with the trajectory on $\widetilde{M}$, which by the Theorem does not converge to any finite cycle.

For the margin, the minimax distributions $u_*^A = (1/5, 2/5, 1/5, 1/5)$ on L_A and $u_*^B = (1/5, 2/5, 1/5, 1/5, 0)$ on L_B each give all four column edges equal to exactly $1/5$. Aggregating these via Lemma 23 produces a distribution D_* on $\widetilde{M}$ under which every positive-column edge equals $\gamma(\widetilde{M}) = 1/5$. Lemma 24 then yields $\gamma(\widetilde{M}') = \gamma(\widetilde{M}) = 1/5$. □

A Appendix

A.1 Additional results

Lemma 26 (Balancing after update [9])**.** *A single AdaBoost update on a selected column $h \in \{\pm 1\}^m$ sends the total post-update mass on the $h = +1$ rows and on the $h = -1$ rows to exactly $1/2$ each.*

Proof. If $\mu = \sum_i w_i h_i$, then the exhaustive AdaBoost update is $w_i' = \frac{w_i}{1+\mu h_i}$. Therefore

$$\sum_{h_i=+1} w_i' = \frac{\sum_{h_i=+1} w_i}{1+\mu} = \frac{(1+\mu)/2}{1+\mu} = \frac{1}{2},$$

and similarly $\sum_{h_i=-1} w_i' = 1/2$. □

A.2 What the certificate proves

The certificate proves all of the following finite assertions.

1. The explicit branch maps $F_0^A, F_1^A, F_0^B, F_1^B$ are derived symbolically from H_0, H_1, and the displayed numerator/denominator polynomials $(P_A, Q_A, D_A, R_A, P_B, Q_B, D_B, R_B, E_B)$ are verified to match the derived rational functions term-for-term.

2. $F_0^B|_{e=0} = F_0^A$, $F_1^B|_{e=0} = F_1^A$, and both B third-coordinate numerators are exactly divisible by e.

3. The 2-cycle equations reduce to $N_1 = N_2 = 0$; the displayed polynomials N_1, N_2 are verified to match the derived expressions term-for-term. Elimination gives the exact degree-7 polynomial f, and the root interval for p_* is unique.

4. The exact polynomial formula for d_* is correct modulo f.

5. The exact elimination polynomial for λ is the irreducible degree-14 polynomial above.

6. The exact elimination polynomial for κ is the irreducible degree-7 polynomial above.

7. The isolating intervals for λ, ν, κ each contain exactly one root.

8. The exact dominant half-step directions v_i^A, v_i^B are certified by explicit rational interval enclosures of $\theta_A, \theta_{B,0}, \chi_B, \theta_{B,1}$.

9. The half-step contractions c_A, c_B are certified and satisfy $c_A < 1$, $c_B < 1$.

10. The explicit sectors S_i^A, S_i^B with $r = 2.5 \times 10^{-8}$ and $\rho = 10^{-3}$ are certified by interval Jacobians and interval Hessian bounds, using the exact first-column structure from Lemma 14.

11. The local branch words H_0, H_1 are certified on the full phase boxes covering those sectors.

12. The whole-burst lock margin is certified.

13. The explicit rational starting point $W_0 = a_0 \otimes b_0$ is certified: its chart image satisfies $\hat{a}_0 \in S_0^A$, $\hat{b}_0 \in S_0^B$.

14. No floating-point number is used in any proof assertion. Floating-point/decimal conversion is used only for human-readable printing of already-certified rational intervals.

15. No negated column is ever selected on $[\widetilde{M} \mid -\widetilde{M}]$: the certificate computes, by exact interval arithmetic on the full sector boxes covering all eight burst/frozen-phase combinations and all five burst steps, interval enclosures of every column edge at every step, and verifies that the selected edge strictly exceeds every negated column edge. The certified minimum gap is $g_{\text{neg}}^{\text{box}} \geq 0.274233539263122735 > 0$.

16. The margin $\gamma([\widetilde{M} \mid -\widetilde{M}]) = 1/5$ is certified: the explicit minimax distributions $u_*^A = (1/5, 2/5, 1/5, 1/5)$ and $u_*^B = (1/5, 2/5, 1/5, 1/5, 0)$ are verified to give all column edges exactly $1/5$ on L_A and L_B respectively, so every negated column edge is $-1/5$.

In the certificate, items 1–14 are verified by `adaboost_cert_core.py`, and items 15–16 are verified by `adaboost_cert_negcols.py`. Running `adaboost_cert_full.py` executes both scripts and prints SHA-256 digests for reproducibility.

B Discussion of collaboration with AI

One of the primary goals of this research was to push the limits of what can be achieved with large language models (LLMs) for novel mathematical research. While many recent efforts have targeted open problems in mathematics, there have been fewer notable results in other areas of the mathematical sciences, such as theoretical computer science. We searched for problems on SolveAll [2]—a curated collection of open research problems spanning machine learning—and selected this problem.

The bulk of the research took roughly two weeks of effort. We began by posing the problem to various frontier language models as a single-shot task. Using the problem statement from SolveAll (shown below), we prompted GPT-5.4 Pro, Claude Opus 4.6, and Gemini Deep Think to solve the problem.

Problem statement (SolveAll)

Let $S = \{(x_i, y_i)\}_{i=1}^m$ be a fixed binary-labeled dataset with $y_i \in \{-1, +1\}$, and let $\mathcal{H}$ be a weak-hypothesis class. Consider the discrete AdaBoost update of Freund & Schapire (1997), in the exhaustive weak-learner regime used in Rudin et al. (2012), with weight vectors $w_t \in \Delta^{m-1}$ (the probability simplex). At iteration t, choose

$$h_t \in \arg\max_{h \in \mathcal{H}} \sum_{i=1}^m w_{t,i}\, y_i\, h(x_i),$$

define weighted error

$$\varepsilon_t = \sum_{i=1}^m w_{t,i}\, \mathbf{1}\{h_t(x_i) \neq y_i\},$$

and update with

$$\alpha_t = \frac{1}{2} \log \frac{1 - \varepsilon_t}{\varepsilon_t}, \qquad w_{t+1,i} = \frac{w_{t,i} \exp(-\alpha_t y_i h_t(x_i))}{Z_t},$$

where Z_t normalizes to $\sum_i w_{t+1,i} = 1$.
Equivalently, with finite hypothesis set $\mathcal{H} = \{\tilde{h}_1, \ldots, \tilde{h}_N\}$ and matrix $M \in \{-1, +1\}^{m \times N}$ defined by $M_{ij} = y_i \tilde{h}_j(x_i)$, step t selects

$$j_t \in \arg\max_{j \in [N]} (w_t^\top M)_j, \qquad h_t = \tilde{h}_{j_t}.$$

As specified in Rudin et al. (2012), if this argmax is not unique, ties are broken in a fixed deterministic way (for concreteness: pick the smallest index j). The generic no-tie condition means the argmax is unique at every iterate, i.e.

$$(w_t^\top M)_j \neq (w_t^\top M)_{j'} \quad \text{for all } j \neq j' \text{ and all } t,$$

equivalently, w_t never lands on a tie boundary between weak-hypothesis regions of the simplex. This induces a discrete dynamical system $T\colon \Delta^{m-1} \to \Delta^{m-1}$ by $w_{t+1} = T(w_t)$.

Unsolved problem target: Does every trajectory converge to a cycle under natural genericity conditions?

It quickly became apparent that the most sophisticated arguments were produced by GPT-5.4 Pro. Opus 4.6 regularly refused to engage because it recognized the problem as open, and Deep Think consistently returned solutions with numerous errors and logical flaws pointed out by both Opus 4.6 and other copies of itself. We therefore focused on generating solution attempts with GPT-5.4 Pro.

Single-shot prompts often ended in partial reductions or failed attempts. However, instead of using the API, we used the web interface to interact with GPT, which more natively supported multi-turn conversations. This allowed us to push the model to try again even if it refused to solve the problem in its first attempt, often by following up with the prompt, "Good progress. However, you must keep pushing to resolve Rudin, Schapire, and Daubechies's problem. The only goal is a correct and complete resolution to the problem."

After many failed attempts—roughly 10 of them—we began to compile previous results from the model into a condensed summary with Claude Opus 4.6, asking it to analyze the various attempts and suggest the most promising directions forward. We then conducted more rounds of summarizing previous attempts and generating new directions to pursue with Claude's recommendations centered on testing additional reductions or simplified cases, none of which proved fruitful when passed back to GPT. However, at around the fifth summarization attempt, Claude produced the following review and list of potential directions.

Summary and proposed directions (Claude Opus 4.6)

Prior work from our research team

We have done extensive work on this problem. Below is a summary of what has been established, organized by its actual usefulness toward a full resolution.

Critical correction: the right margin condition. One collaborator identified that much of our earlier work used the wrong margin quantity. The quantity $\rho(M) = \max_{d\in\Delta^{m-1}} \min_{j\in[N]}(M^\top d)_j$ (which asks for a distribution over *examples* making all hypothesis edges positive) is **not** the relevant weak-learning margin. The correct quantity is the minimax game value:

$$\gamma(M) = \min_{w\in\Delta^{m-1}} \max_{j\in[N]}(w^\top M)_j = \max_{\lambda\in\Delta^{N-1}} \min_{i\in[m]}(M\lambda)_i.$$

This is the quantity guaranteeing AdaBoost's chosen edge is uniformly positive. One can make $\rho > 0$ for any matrix by appending an all-positive row, so "$\rho > 0$" does not isolate the hard case. Any proof strategy must use $\gamma(M) > 0$.

Useful: clarification that "eventual periodicity" is the wrong target. An explicit counterexample to *literal* eventual periodicity was constructed. Consider 5 hypotheses with negative-set family $\{0,2\},\{1,2\},\{2,3\},\{2,4\},\{0,1,3\}$, giving margin $\rho = 0$. On a carefully constructed invariant region, generic no-tie orbits satisfy $q_n \sim 9/(16n) \to 0$—they converge to a boundary fixed point but never literally repeat. Similarly, for $N = 3$ under the genuine weak-learning margin, if an all-positive row $(+1,+1,+1)$ carries initial mass $d_0 > 0$, then $d_t \to 0$ exponentially but never reaches zero, so the orbit never literally enters a finite cycle even though it converges phasewise to a 3-cycle.
Consequence: The right conjecture is *phasewise convergence to a cycle* (i.e., convergence of the weight vector to a periodic orbit after quotienting out vanishing components), not literal eventual entry into a repeating sequence of weight vectors.

Useful: exact reformulation as coordinate descent. After aggregating examples by their sign pattern $r = (M_{i1}, \ldots, M_{iN}) \in \{-1, +1\}^N$, with aggregate mass $w_t(r)$, the dynamics takes the exponential-family form:

$$w_t(r) = \frac{q_r e^{-\lambda_t \cdot r}}{\sum_s q_s e^{-\lambda_t \cdot s}}, \qquad \lambda_{t+1} = \lambda_t + \operatorname{artanh}(\mu_{j_t}(\lambda_t))\, e_{j_t},$$

where $\mu_j(\lambda) = \sum_r w_\lambda(r)\, r_j$ is the edge of hypothesis j, and $j_t \in \arg\max_j \mu_j(\lambda_t)$.
This is **greedy (Gauss–Southwell) coordinate descent** on the convex potential

$$F(\lambda) = \log \sum_r q_r e^{-\lambda \cdot r}.$$

Useful: toric invariant structure. Let $C \in \{0, 1\}^{(N+1)\times q}$ be the extended incidence matrix (first row all-ones, remaining rows recording which hypotheses each sign-pattern gets wrong). For every $z \in \ker C$, the monomial $I_z(x) = \prod_\ell x_\ell^{z_\ell}$ is exactly invariant under every branch map T_j. So orbits are confined to toric leaves. This is an exact structural fact, not an approximation.

Useful: periodic orbits carry canonical equal-weight vectors. If x_0 lies on a period-p orbit with step sizes β_t, define cumulative beta $\Gamma_j = \sum_{t:j_t=j} \beta_t$. Then $\sum_{j\in A} \Gamma_j$ is constant across all support blocks A. So every periodic orbit determines a positive vector in an "equal-weight cone"—the beta-weighted visit frequencies must balance across all active support sets.

Correct but low strategic value: $N = 3$ theorem. Under the genuine weak-learning margin $\gamma(M) > 0$ plus nondegeneracy, for $N = 3$ the only possible sign patterns are the three singleton-negative rows plus an optional all-positive row. Errors simplify to $\varepsilon_j = x_j$, the itinerary locks to cyclic order from round 1, and the return map is projective-linear. Perron–Frobenius gives phasewise convergence to a 3-cycle. This is a complete theorem, but the proof mechanism (everything becomes linear) is entirely specific to $N = 3$ and gives no leverage on $N \geq 4$.

Not useful toward a resolution: $N = 4$ classification. The team exhaustively classified all 13 feasible active families at $N = 4$ (up to relabeling), found 5 that cover all 4 coordinates, and showed only one is irreducible and multi-negative: $\{4, 12, 13, 23\}$. For that family, an explicit 2D return map in sorted ratio coordinates (P, R) was analyzed: transient branches B, C were shown to increase R, and the eventual regime is either A^∞ (converging to a 6-cycle) or $(DA)^\infty$ (converging to a 10-cycle, proved via contraction with $\|DH\|_\infty \leq 0.65$).
This is mathematically correct but **strategically a dead end**: you cannot close the problem by classifying one N at a time when the number of families and possible cycle periods both grow without bound (periods up to 219 were observed at $N = 7$).

Not useful: $N = 5$ enumeration and computational scans. 47 candidate families were enumerated at $N = 5$, with 22 numerically retaining all 5 coordinates. All tested orbits cycled. Six distinct asymptotic direction patterns and periods $\{5, 6, 7, 8, 9, 10, 12, 14\}$ were observed. This is computational evidence only and provides no path to proof.

Not useful: repeated reformulations. Across the prior work, the dynamics was restated as: KL projection / Gauss–Southwell, weighted-homogeneous quotient maps, greedy birational systems on antichain hypergraphs, toric-leaf dynamics, multiaffine polynomial replacement. These are all the same object in different coordinates. None introduces a genuinely new proof technique.

What is needed: a top-down argument for all N

The prior work has been almost entirely bottom-up case analysis. The problem almost certainly requires a structural argument that works at all N simultaneously. Here are the most promising directions, in decreasing order of tractability:

Direction 1: Exploit the coordinate-descent / potential structure. AdaBoost is greedy coordinate descent on $F(\lambda) = \log \sum_r q_r e^{-\lambda \cdot r}$, which is convex and real-analytic. The step sizes are exact: $\Delta\lambda_j = \operatorname{artanh}(\mu_j(\lambda))$. This is not generic coordinate descent—the step size is the *exact* one-dimensional minimizer along each coordinate axis.
Key properties to exploit:

- F decreases by $-\log(2\sqrt{\varepsilon_t(1-\varepsilon_t)})$ at each step, which is bounded below by $-\log(1-\gamma^2/2) > 0$ when $\gamma(M) > 0$.
- The gradient $\nabla F(\lambda) = -\mu(\lambda)$, where $\mu_j(\lambda)$ is the edge of hypothesis j under w_λ.
- After each exact coordinate minimization along axis j, the new edge satisfies $\mu_j(\lambda^+) = 0$—the selected coordinate's gradient is zeroed out.
- The sublevel sets of F are compact (since $\gamma > 0$ implies $F(\lambda) \to +\infty$ as $\|\lambda\| \to \infty$... actually, F is bounded below and its sublevel sets need careful analysis).

The key question: for exact coordinate descent on a real-analytic convex function where each step zeros out one partial derivative, can you prove eventual periodicity of the coordinate selection sequence?
This connects to the literature on cyclic and greedy coordinate descent. Relevant results: Beck and Tetruashvili (2013) on convergence rates, Nutini et al. (2015) on greedy coordinate descent, and the general theory of nonlinear Gauss–Seidel methods. The specific structure here (exact minimization, real-analyticity, finite partition of the domain into greedy-selection regions) may admit a proof via the Łojasiewicz inequality or related real-analytic geometry.

Direction 2: Real-analytic dynamics and the branch partition. The simplex is partitioned into finitely many open regions $\Omega_j = \{w : (w^\top M)_j > (w^\top M)_k \ \forall k \neq j\}$, and on each region T is a fixed real-analytic map T_j. The question is whether the symbolic itinerary $j_0, j_1, j_2, \ldots$ is eventually periodic.
After quotienting by the toric invariants, orbits live on a compact real-analytic manifold of dimension at most $N-1$. The return map on each branch is a rational map. The set of accumulation points of a non-periodic orbit would be a compact invariant set, and its structure is constrained by the real-analyticity of each branch.
A possible approach: show that the ω-limit set of any orbit is a finite union of periodic orbits. For piecewise real-analytic maps, this might follow from showing that the orbit can only visit finitely many distinct branch sequences (using that each branch map is a contraction in some metric, or that the potential F provides a strict Lyapunov function for the branch dynamics).

Direction 3: Contraction of composite return maps. In the $N = 4$ case, the team found that the two-step return map $H = T_A \circ T_D$ is a strict contraction ($\|DH\|_\infty \leq 0.65$). Is this an accident of low dimension, or is there a general mechanism?
For any eventually periodic branch word $\sigma = (j_0, \ldots, j_{p-1})$, the return map is $R_\sigma = T_{j_{p-1}} \circ \cdots \circ T_{j_0}$. Each T_j maps the selected coordinate's error to exactly $1/2$ and rescales everything else. The conjecture that all such return maps are eventually contracting (in some norm, on the relevant toric leaf) would immediately imply convergence to a unique periodic orbit for each branch word, and then finiteness of realizable branch words would give eventual periodicity.

Direction 4: The $\mu_j = 0$ post-condition as a reset mechanism. After selecting hypothesis j, the new weight vector satisfies $\mu_j(\lambda^+) = 0$, i.e., hypothesis j has zero edge. This is a strong constraint: it means the orbit is always being projected onto a coordinate hyperplane of the edge space. The itinerary question becomes: starting from a point where $\mu_j = 0$, which coordinate k has the largest edge, and after zeroing μ_k, does the sequence of zeroed coordinates eventually repeat?
This "reset" structure—each step annihilates one coordinate of the gradient—is very specific and may admit a combinatorial/topological argument about the sequence of active constraints.

Your task. You have the full problem statement, the correct margin condition ($\gamma(M) > 0$), the known structural facts (coordinate descent on F, toric invariants, $\mu_j = 0$ post-condition, canonical beta vectors on periodic orbits), and four promising directions for a general proof. Focus on whichever direction you find most tractable. Be careful, precise, and rigorous. We need a complete resolution. Remember, the only thing we care about is producing a novel and correct solution to the problem.

GPT then provided the following response that, while not a solution, led Claude to propose pursuing a disproof via a counterexample.

Structural reduction of the cycling problem (GPT-5.4 Pro)

The Rudin–Daubechies–Schapire papers already show that AdaBoost admits genuine low-dimensional cyclic dynamics, and the COLT open problem asks for a full characterization of whether the weight distributions converge to a cycle in general.
I do not have a certified full resolution of the general problem. What I can certify is a sharp structural reduction. It isolates the real obstruction and shows exactly what a complete proof or disproof now has to control.

1. The compact analytic-descent route cannot work. Let

$$F(\lambda) = \log \sum_r q_r e^{-\lambda \cdot r}, \qquad \mu_t = \max_j \mu_j(\lambda_t), \qquad \beta_t = \operatorname{artanh}(\mu_t).$$

Under the correct weak-learning margin assumption $\gamma(M) > 0$, every selected edge satisfies

$$\mu_t \geq \gamma > 0,$$

hence

$$\beta_t \geq \beta_* := \operatorname{artanh}(\gamma) > 0.$$

Because AdaBoost performs the exact coordinate minimizer,

$$F(\lambda_{t+1}) - F(\lambda_t) = \log Z_t = \tfrac{1}{2}\log(1-\mu_t^2) \le \tfrac{1}{2}\log(1-\gamma^2) < 0.$$

Therefore

$$F(\lambda_t) \le F(\lambda_0) + t \cdot \tfrac{1}{2}\log(1-\gamma^2) \to -\infty$$

linearly, and also

$$\|\lambda_t\|_1 \ge t\,\beta_*.$$

So the orbit does **not** remain in a compact sublevel set of F. Any proof strategy based on ordinary real-analytic coordinate-descent convergence to a stationary point is therefore mis-aimed. The asymptotic object is necessarily a drift to infinity with a finite-dimensional transverse dynamics riding on top of it.
That is the first rigorous conclusion: the problem lives at infinity, not in a compact analytic basin.

2. AdaBoost admits an exact asynchronous product decomposition. This is the clean top-down structure.
Suppose the sign-pattern space splits as a product

$$\mathcal{R} = \mathcal{R}_A \times \mathcal{R}_B$$

and the base masses factor as

$$q(r_A, r_B) = q_A(r_A)\, q_B(r_B).$$

Write $\lambda = (\lambda^A, \lambda^B)$. Then

$$F(\lambda^A, \lambda^B) = \log \sum_{r_A, r_B} q_A(r_A)\, q_B(r_B)\, e^{-\lambda^A \cdot r_A - \lambda^B \cdot r_B} = F_A(\lambda^A) + F_B(\lambda^B).$$

Consequently

$$w_\lambda(r_A, r_B) = w^A_{\lambda^A}(r_A)\, w^B_{\lambda^B}(r_B),$$

and every edge in block A depends only on λ^A, every edge in block B only on λ^B.
Hence the full AdaBoost dynamics on the product system is exactly:

- compute the current best local edge in subsystem A,
- compute the current best local edge in subsystem B,
- update only the subsystem whose local best edge is larger.

So global AdaBoost is closed under **asynchronous competition of local AdaBoost subsystems**.
This is the key reduction. Any universal theorem must survive such products. Any counterexample can be built by finding two local subsystems whose competition produces a nonperiodic symbolic schedule.

3. The correct local invariant near a periodic orbit. Now fix a local subsystem with a period-p orbit

$$x_0^*,\, x_1^*,\, \ldots,\, x_{p-1}^*,$$

selected coordinates $j_0, \ldots, j_{p-1}$, step sizes $\beta_0, \ldots, \beta_{p-1}$, and define

$$\Gamma := \sum_{t=0}^{p-1} \beta_t e_{j_t} \in \mathbb{R}^N.$$

For any active sign pattern r on the cycle support, the exact p-step update gives

$$x_{t+p}(r) = x_t(r) \frac{e^{-r\cdot\Gamma}}{\prod_{\tau=0}^{p-1} Z_\tau}.$$

Since $x_{t+p}(r) = x_t(r)$ on the cycle, every active r satisfies

$$r \cdot \Gamma = c, \qquad c := -\sum_{\tau=0}^{p-1} \log Z_\tau.$$

This recovers the equal-weight relation in the strongest useful form: the whole cycle is encoded by the single vector Γ, and the active support lies on the affine hyperplane $r \cdot \Gamma = c$.
Now append one extra sign pattern s with tiny mass u. Along the affine slice

$$x_t(u) = (1-u)\, x_t^* + u\, \delta_s,$$

the current edge of coordinate j is exactly

$$\mu_{t,j}(u) = (1-u)\, \mu_{t,j}^* + u\, s_j = \mu_{t,j}^* + (s_j - \mu_{t,j}^*)\, u.$$

In particular, for the selected coordinate j_t,

$$\mu_t(u) = \mu_t^* + (s_{j_t} - \mu_t^*)\, u.$$

After one full period, the extra mass is multiplied by

$$u' = e^{c - s\cdot\Gamma}\, u + O(u^2).$$

So every extra sign pattern produces a scalar transverse mode whose:

- instantaneous score correction is $s_{j_t} - \mu_t^*$,
- full-period contraction exponent is $c - s \cdot \Gamma$.

If the visible cycle is hyperbolically attracting on its toric leaf, this scalar mode persists on an analytic invariant graph.
This is the exact local normal-form data. It is the right replacement for generic "coordinate descent on an analytic function."

4. The exact normal form that would disprove phasewise cycling. Here is the decisive reduction.
Assume we have two factorized local gadgets A, B, each near a phase-synchronized local cycle, with scalar mode coordinates $u, v > 0$, and suppose on the chosen section their current best edges are

$$E_A = \bar{\mu} + a_A u, \qquad E_B = \bar{\mu} + a_B v, \qquad a_A, a_B > 0,$$

while selecting A multiplies u by $\kappa_A \in (0,1)$ and leaves v unchanged, and selecting B multiplies v by $\kappa_B \in (0,1)$ and leaves u unchanged.
Set

$$x = \log \frac{a_A u}{a_B v}, \qquad \alpha = -\log \kappa_A, \qquad \beta = -\log \kappa_B.$$

Choose the canonical representative $x \in [-\beta, \alpha)$. Then the branch rule is:

$$x \mapsto \begin{cases} x - \alpha, & x \in [0, \alpha), \\ x + \beta, & x \in [-\beta, 0). \end{cases}$$

This map is exactly conjugate to rigid rotation by β on a circle of length $\alpha + \beta$.
Therefore:

- if $\beta/(\alpha+\beta) \in \mathbb{Q}$, the symbolic itinerary is periodic;
- if $\beta/(\alpha+\beta) \notin \mathbb{Q}$, the symbolic itinerary is Sturmian and **not eventually periodic**.

So a complete disproof of universal phasewise cycling reduces to finding one genuine AdaBoost realization of this scalar two-mode normal form with irrational rotation number.
That is a rigorous theorem, not a heuristic.

5. Why the simplest counterexample mechanism does not work. There is also a rigorous obstruction.
Suppose a scalar mode comes from appending a passive sign pattern s that is correct on **every** hypothesis actually visited on the visible cycle, i.e.

$$s_j = +1 \quad \text{for all } j \text{ with } \Gamma_j > 0.$$

Then

$$s \cdot \Gamma = \sum_{\Gamma_j > 0} \Gamma_j,$$

so its contraction factor is

$$\kappa_s = e^{c - s\cdot\Gamma} = e^{c - \sum_{\Gamma_j > 0} \Gamma_j},$$

which is independent of s.
Thus two "always-correct passive rows" can never supply two distinct exponents $\kappa_A \neq \kappa_B$.
The naive plan

- take a constant-edge local cycle,
- append two harmless always-correct rows,
- form a product,

- get irrational rotation,

cannot succeed.
This is why the obvious symmetric examples do not immediately generate a counterexample. Any disproof has to use either sign-changing passive modes or intrinsic stable modes of the visible cycle itself.

6. What this reduces the original problem to. The problem is no longer "analyze all global orbits directly." It reduces to a local spectral classification problem at infinity.
A full **negative** solution would follow from one explicit hyperbolic local cycle with two scalar stable modes whose competition realizes the rotation normal form above.
A full **positive** solution would have to prove that no such mode pair can ever occur in any factorized product of hyperbolic local cycles.
So the correct top-down statement is:

> *The AdaBoost cycling problem is equivalent to ruling in or ruling out irrational circle-rotation codings generated by stable transverse modes of local cycles in asynchronous product systems.*

That is the strongest rigorous reduction I can certify from the material supplied plus the structural calculations above.
The unresolved step is explicit realization or universal exclusion of that rotation normal form inside the actual AdaBoost class. The next mathematically sharp target is to classify constant-edge hyperbolic cycles and their stable scalar modes; that is the narrow bottleneck now.

After being asked to provide another summary, Claude returned the analysis below and then posed two possible next steps:

Summary (Claude Opus 4.6)

Prior work from our research team We have done extensive work. Below is everything established so far, filtered to what is genuinely useful.
1. The correct margin condition The relevant weak-learning margin is:

$$\gamma(M) = \min_{w \in \Delta^{m-1}} \max_{j \in [N]} (w^\top M)_j = \max_{\lambda \in \Delta^{N-1}} \min_{i \in [m]} (M\lambda)_i.$$

Some of our earlier work incorrectly used $\rho(M) = \max_{d \in \Delta^{m-1}} \min_{j \in [N]} (M^\top d)_j$, which can be made positive for any matrix by appending an all-positive row and therefore does not isolate the hard case. Everything below uses $\gamma(M) > 0$.
2. The right target is phasewise convergence, not literal periodicity An explicit counterexample to *literal* eventual periodicity exists: a $\rho = 0$ system on 5 hypotheses where generic no-tie orbits satisfy $q_n \sim 9/(16n) \to 0$, converging to a boundary fixed point but never repeating. Similarly, for $N = 3$ with an all-positive row carrying mass $d_0 > 0$, $d_t \to 0$ exponentially but never reaches zero. So the conjecture should be understood as phasewise convergence to a cycle, not literal entry into a finite repeating sequence.

3. Exact reformulation as coordinate descent After aggregating examples by sign pattern $r = (M_{i1}, \ldots, M_{iN}) \in \{-1, +1\}^N$ with aggregate mass $w_t(r)$, the dynamics is:

$$w_t(r) = \frac{q_r e^{-\lambda_t \cdot r}}{\sum_s q_s e^{-\lambda_t \cdot s}}, \qquad \lambda_{t+1} = \lambda_t + \operatorname{artanh}(\mu_{j_t}(\lambda_t))\, e_{j_t},$$

where $\mu_j(\lambda) = \sum_r w_\lambda(r)\, r_j$ is the edge of hypothesis j, and $j_t \in \arg\max_j \mu_j(\lambda_t)$.
This is **greedy (Gauss–Southwell) coordinate descent** on the convex potential:

$$F(\lambda) = \log \sum_r q_r e^{-\lambda \cdot r}.$$

4. The problem lives at infinity, not in a compact basin Under $\gamma(M) > 0$, the selected edge satisfies $\mu_t \geq \gamma > 0$, so $\beta_t \geq \operatorname{artanh}(\gamma) > 0$. The potential decreases linearly: $F(\lambda_t) \leq F(\lambda_0) + \frac{t}{2}\log(1-\gamma^2) \to -\infty$. Also $\|\lambda_t\|_1 \geq t \cdot \operatorname{artanh}(\gamma) \to \infty$. **This rules out any proof strategy based on compactness of sublevel sets or convergence to a stationary point.** The asymptotic dynamics is drift to infinity with a finite-dimensional transverse component.
5. Toric invariant structure Let $C \in \{0,1\}^{(N+1)\times q}$ be the extended incidence matrix. For every $z \in \ker C$, the monomial $I_z(x) = \prod_\ell x_\ell^{z_\ell}$ is exactly invariant under every branch map T_j. Orbits are confined to toric leaves, reducing the effective dimension to at most $N-1$.
6. Periodic orbits carry canonical equal-weight vectors On a period-p orbit with step sizes β_t and selected coordinates j_t, define cumulative beta $\Gamma = \sum_{t=0}^{p-1} \beta_t e_{j_t}$. Then every active sign pattern r satisfies $r \cdot \Gamma = c$ where $c = -\sum_\tau \log Z_\tau$. This is the exact constraint that the beta-weighted visit frequencies must balance across all support sets.
7. Asynchronous product decomposition If the sign-pattern space factors as $\mathcal{R} = \mathcal{R}_A \times \mathcal{R}_B$ with $q(r_A, r_B) = q_A(r_A) q_B(r_B)$, then $F(\lambda^A, \lambda^B) = F_A(\lambda^A) + F_B(\lambda^B)$ and the full AdaBoost dynamics is: compute the best edge in each subsystem, update whichever subsystem has the larger best edge. Global AdaBoost decomposes as **asynchronous competition between independent local AdaBoost subsystems**. Any universal theorem must survive such products.
8. The decisive structural finding: a circle-rotation normal form This is the most important observation across all our work. Consider two factorized local subsystems A, B, each near a hyperbolic periodic orbit, with scalar stable transverse modes parameterized by $u, v > 0$. Suppose on the relevant Poincaré section their current best edges are:

$$E_A = \bar{\mu} + a_A u, \qquad E_B = \bar{\mu} + a_B v, \qquad a_A, a_B > 0,$$

and selecting A contracts u by $\kappa_A \in (0,1)$ (leaving v fixed), while selecting B contracts v by $\kappa_B \in (0,1)$ (leaving u fixed).
Setting $x = \log(a_A u / a_B v)$, $\alpha = -\log \kappa_A$, $\beta = -\log \kappa_B$, the branch rule on the circle $\mathbb{R}/(\alpha+\beta)\mathbb{Z}$ is:

$$x \mapsto \begin{cases} x - \alpha, & x \geq 0, \\ x + \beta, & x < 0. \end{cases}$$

This is exactly conjugate to **rigid rotation by β on a circle of circumference** $\alpha + \beta$. Therefore:

- If $\beta/(\alpha+\beta) \in \mathbb{Q}$, the symbolic itinerary is eventually periodic. **The conjecture holds.**
- If $\beta/(\alpha+\beta) \notin \mathbb{Q}$, the symbolic itinerary is Sturmian and **never eventually periodic. The conjecture is false.**

9. The simplest counterexample attempt fails (but the mechanism is not dead) If two passive sign patterns are both correct on every hypothesis visited by the visible cycle ($s_j = +1$ for all j with $\Gamma_j > 0$), then they both have the same contraction factor $\kappa_s = e^{c - \sum_{\Gamma_j > 0} \Gamma_j}$, so $\alpha = \beta$ and $\beta/(\alpha + \beta) = 1/2 \in \mathbb{Q}$. The simplest symmetric construction always gives rational rotation number.
This does not kill the mechanism. It means a counterexample requires either sign-changing passive modes (patterns with some $s_j = -1$ on visited coordinates, producing distinct contraction rates) or intrinsic stable modes of the visible cycle itself.

10. Low-dimensional results (limited strategic value) $N = 3$: fully resolved under $\gamma > 0$. Proof is completely specific to $N = 3$ (everything becomes linear). $N = 4$: all active families classified, all eventually periodic by explicit return map analysis. Cannot scale. Computational scans through $N = 7$ always show eventual cycles (periods up to 219), but this is not a proof.

There are two paths:

Path A (Disproof): Construct an explicit M-matrix that decomposes as a product of two subsystems, each admitting a hyperbolic periodic orbit with a scalar stable mode, where the contraction exponents $\alpha = -\log \kappa_A$ and $\beta = -\log \kappa_B$ satisfy $\beta/(\alpha+\beta) \notin \mathbb{Q}$. The contraction exponent of a passive sign pattern s appended to a local cycle with cumulative beta vector Γ is $\kappa_s = e^{c - s \cdot \Gamma}$. The simplest case (all-positive s) forces $\alpha = \beta$, so you need sign-changing passive modes: patterns with some $s_j = -1$ on coordinates visited by the cycle, giving $s \cdot \Gamma \neq s' \cdot \Gamma$ for two distinct patterns s, s'. Then $\log \kappa_A / \log \kappa_B = (c - s \cdot \Gamma)/(c - s' \cdot \Gamma)$, and you need this to be irrational. Since c and Γ involve artanh of rational functions composed through cycle steps, irrationality of their ratio is plausible but must be verified for a concrete construction.

Path B (Proof): Prove that AdaBoost's structure forces $\log \kappa_A / \log \kappa_B \in \mathbb{Q}$ universally. This would require showing that contraction exponents arising from the specific algebraic form of AdaBoost ($\beta_t = \operatorname{artanh}(\mu_t)$, μ_t rational in the weights, toric invariants constraining eigenvalues of return maps) always maintain rational log-ratios. Note: a universal rationality constraint on eigenvalue ratios of piecewise-rational return maps would be remarkable and is likely false in general, which is circumstantial evidence favoring Path A.

Focus on whichever direction you find most tractable. Be careful, precise, and rigorous. We need a complete resolution. (This italicized section was added by the authors.)

GPT's first few attempts each pursued the proof direction (Path B), but it eventually attempted to provide a disproof via the counterexample suggested in Path A. After several failed counterexample attempts (an example of which is shown in B), GPT provided the following solution, which served as the foundation for the final solution detailed in the main text.

GPT-5.4 Pro's First Successful Counterexample Attempt

Theorem. There exists a finite $\{-1, +1\}$-valued matrix $\widetilde{M}$ with weak-learning margin $\gamma(\widetilde{M}) = 1/5 > 0$ such that standard exhaustive AdaBoost on $\widetilde{M}$, started from the uniform distribution and using smallest-index tie-breaking, does **not** converge to any finite cycle.

Therefore, the COLT question regarding whether AdaBoost always cycles has a **negative** answer.

1. The Two Local Gadgets Define the following matrices:

$$L_A = \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad L_B = \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

Let $M_0 = L_A \boxplus L_B$ be the 20×8 block product:

$$(M_0)_{(p,q),j} = (L_A)_{p,j} \quad (1 \le j \le 4), \qquad (M_0)_{(p,q),4+j} = (L_B)_{q,j} \quad (1 \le j \le 4)$$

For any joint distribution W on the 20 rows, write its marginals as $a \in \Delta^3$ on the L_A-rows and $b \in \Delta^4$ on the L_B-rows. Then the edge vector splits cleanly:

$$(W^\top M_0)_j = (a^\top L_A)_j \quad (1 \le j \le 4), \qquad (W^\top M_0)_{4+j} = (b^\top L_B)_j \quad (1 \le j \le 4)$$

Hence, on product states $W = a \otimes b$, selecting an A-column updates only a, and selecting a B-column updates only b.

2. Weak-Learning Margin Let $\beta = (2, 1, 1, 1)$. Every row r of L_A satisfies $r \cdot \beta = 1$, and every row r of L_B satisfies $r \cdot \beta \in \{1, 3\}$. Therefore, for any distribution u on L_A or L_B:

$$2\mu_1 + \mu_2 + \mu_3 + \mu_4 = \mathbb{E}[r \cdot \beta] \ge 1$$

Thus $\max_j \mu_j \ge 1/5$, implying $\gamma(L_A) \ge 1/5$ and $\gamma(L_B) \ge 1/5$.
Equality holds for L_A at $u_* = (1/5, 2/5, 1/5, 1/5)$, since $u_*^\top L_A = (1/5, 1/5, 1/5, 1/5)$. Because L_A sits inside L_B as the first four rows, $\gamma(L_B) \le 1/5$ as well. Hence:

$$\gamma(L_A) = \gamma(L_B) = 1/5$$

For the product matrix M_0:

$$\gamma(M_0) = \min_{a,b} \max \Big(\max_j (a^\top L_A)_j,\ \max_j (b^\top L_B)_j \Big) = \max(\gamma(L_A), \gamma(L_B)) = 1/5$$

3. The Common Period-10 Orbit Both L_A and L_B have the same active period-10 orbit on the face $w_5 = 0$, governed by the 5-step half-words:

$$H_0 = (1, 3, 4, 1, 2), \qquad H_1 = (1, 4, 3, 1, 2)$$

The cycle points at the start of these half-words are:

$$P_0 = (0.1371168\ldots,\ 0.3628831\ldots,\ 0.2420046\ldots,\ 0.2579953\ldots)$$
$$P_1 = (0.1371168\ldots,\ 0.3628831\ldots,\ 0.2579953\ldots,\ 0.2420046\ldots)$$

At both P_0 and P_1, the selected minimum edge is $\mu_* = 0.2742336\ldots$ for hypothesis 1.
Along each half-word, the other four selected edges are strictly larger. The minimum gap between the selected edge and the best unselected edge is $\Delta \approx 0.0758695 > 0$.

Additionally, we define the local branch gap $g_{\mathrm{br}} > 0$ as the minimum, over all 10 orbit points in both gadgets and both half-words, of the selected edge minus the largest competing edge inside the same gadget. Numerically, $g_{\mathrm{br}} \approx 0.0250986 > 0$. This ensures the local 5-step branch maps are genuinely analytic on neighborhoods of P_i.

4. The Two Competing Stable Multipliers For L_A, the period-10 return at P_0 has a dominant stable multiplier $\lambda \approx 0.0944920\ldots$ and smaller stable multipliers $0.0011453\ldots$ and 0.

For L_B, the period-10 return at P_0 has a dominant stable multiplier $\kappa \approx 0.2038817\ldots$ and smaller stable multipliers λ, $0.0011453\ldots$, and 0.

The key arithmetic fact is that their logarithms are incommensurate:

$$\frac{\log \lambda}{\log \kappa} \notin \mathbb{Q}$$

Proof: The multiplier λ is a root of an irreducible degree-14 polynomial over $\mathbb{Q}$ with leading coefficient 16 and constant term 4, yielding algebraic norm $N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda) = 1/4$. The multiplier κ is a root of the irreducible polynomial $z^7 - 35z^6 + 385z^5 - 1835z^4 + 2315z^3 - 625z^2 + 51z - 1$, yielding $N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa) = 1$. If $\log \lambda / \log \kappa \in \mathbb{Q}$, then $\lambda^m = \kappa^n$ for integers $m, n > 0$. Taking norms in the compositum field forces $(1/4)^{m[L:\mathbb{Q}(\lambda)]} = 1$, which is impossible.

5. Adapted Dominant Coordinates and Invariant Cones Fix $G \in \{A, B\}$. Let F_i^G be the analytic 5-step branch map on the half-word H_i, sending a neighborhood of P_i^G to one of $P_{i\oplus 1}^G$. Let $J_i^G = DF_i^G(P_i^G)$.

The dominant full-return multipliers are $\mu_A = \lambda$ and $\mu_B = \kappa$. Choose $\chi_i^G > 0$ such that $\chi_0^G \chi_1^G = \mu_G$. We extract the right and left dominant eigenvectors e_i^G and ϕ_i^G so that:

$$J_i^G e_i^G = \chi_i^G e_{i\oplus 1}^G, \qquad \phi_{i\oplus 1}^G J_i^G = \chi_i^G \phi_i^G$$

We normalize these vectors by $\phi_i^G(e_i^G) = 1$ and $D(E - \mu_*)(P_i^G)(e_i^G) = 1$. The second normalization is guaranteed by verified positive visibility.

Define the transverse spaces $N_i^G = \ker \phi_i^G$. The Jacobian restricts to these spaces: $J_i^G N_i^G \subset N_{i\oplus 1}^G$. Let A_i^G be this restriction. Because the transverse spectrum is strictly bounded by the dominant multipliers in both gadgets, we have $\rho(A_1^G A_0^G) < \chi_0^G \chi_1^G$.

Adapted Norms: By standard linear algebra, if $\rho(A_1 A_0) < \tau_0 \tau_1$, there exist equivalent norms $|\cdot|_0$ and $|\cdot|_1$ such that $|A_0 y|_1 \le \tau_0 |y|_0$ and $|A_1 z|_0 \le \tau_1 |z|_1$. We choose $\tau_i^G < \chi_i^G$ and equip N_i^G with such adapted norms $|\cdot|_{i,G}$.

Any point q near P_i^G uniquely decomposes as $q = P_i^G + x e_i^G + y$, with $y \in N_i^G$ and $x = \phi_i^G(q - P_i^G)$. Taylor expansion yields:

$$x^+ = \chi_i^G x + O((|x| + |y|)^2)$$
$$y^+ = A_i^G y + O((|x| + |y|)^2)$$

Let $d_i^G(q) = E(q) - \mu_*$ be the actual selected-edge excess. Since $d_i^G(P_i^G) = 0$ and $Dd_i^G(e_i^G) = 1$, we have $d_i^G(q) = x + \ell_i^G(y) + O((|x| + |y|)^2)$ for some linear form ℓ_i^G.

Fix $\eta, r > 0$ small and define the one-sided cone sectors:

$$S_i^G = \left\{ P_i^G + x e_i^G + y : 0 < x < r,\ |y|_{i,G} \le \eta x \right\}$$

Because $\tau_i^G < \chi_i^G$, shrinking η and r ensures the cones are forward-invariant: $F_i^G(S_i^G) \subset S_{i\oplus 1}^G$. Furthermore, the transverse ratio contracts whenever the gadget updates. Consequently, for

any prescribed $\delta > 0$, the true defect strongly aligns with the pure coordinate x:

$$(1-\delta)x \le d_i^G(q) \le (1+\delta)x \qquad (q \in S_i^G)$$

6. Bursts on the Product Manifold We work on the invariant product manifold $\mathcal{P} = \{a \otimes b : \ a \in \Delta^3, \ b \in \Delta^4\}$. At a burst boundary $W_{5n} = a_n \otimes b_n$, we assume $a_n \in S_{i_n}^A$ and $b_n \in S_{j_n}^B$.
Write their local coordinates as:

$$a_n = P_{i_n}^A + x_n e_{i_n}^A + y_n, \qquad b_n = P_{j_n}^B + s_n e_{j_n}^B + z_n$$

The best current A-edge is $\mu_* + d_{i_n}^A(a_n)$ and the best B-edge is $\mu_* + d_{j_n}^B(b_n)$. Global tie-breaking favors A iff $d_{i_n}^A(a_n) \ge d_{j_n}^B(b_n)$.
Define the ambiguity ratio $\vartheta = (1+\delta)/(1-\delta)$. By the comparability estimate:

- If $x_n \ge \vartheta s_n$, A is forced to win.
- If $x_n < \vartheta^{-1} s_n$, B is forced to win.

Only the narrow strip $\vartheta^{-1} \le x_n/s_n < \vartheta$ is ambiguous.
Shrink the sectors so that every edge changes by at most $\min(g_{\mathrm{br}}, \Delta)/4$ inside them. If A wins the first round, its local itinerary is exactly H_{i_n}. After one A-update, its selected edge is at least $\mu_* + \Delta/2$, while the frozen B-edge remains $\mu_* + O(s_n)$. Thus, the first round firmly dictates the entire 5-step burst.

7. Bounded Log-Ratio and Irrational Winner Frequency Define the pure log-ratio $\ell_n = \log(x_n/s_n)$.
If A wins burst n, B is frozen. The state updates as $x_{n+1} = \chi_{i_n}^A x_n(1+\varepsilon_n)$ and $s_{n+1} = s_n$. Because the trajectory remains in the cone, the Taylor remainder gives $|\varepsilon_n| \le C x_n$. Likewise, if B wins, $s_{n+1} = \chi_{j_n}^B s_n(1+\varepsilon_n)$ with $|\varepsilon_n| \le C s_n$.
Set $\alpha_i = -\log \chi_i^A > 0$ and $\beta_j = -\log \chi_j^B > 0$. The increment equation is:

$$\ell_{n+1} = \begin{cases} \ell_n - \alpha_{i_n} + r_n, & A \text{ wins} \\ \ell_n + \beta_{j_n} + r_n, & B \text{ wins} \end{cases}$$

where $r_n = \log(1+\varepsilon_n)$, satisfying $|r_n| \le C'(x_n + s_n)$. By taking the sector size r small enough, we ensure $|r_n| \le \frac{1}{4}\min(\alpha_{\min}, \beta_{\min})$.
Let $c = \log \vartheta$.

- If $\ell_n \ge c$, A is forced to win, yielding $\ell_{n+1} \le \ell_n - \frac{1}{2}\alpha_{\min}$.
- If $\ell_n \le -c$, B is forced to win, yielding $\ell_{n+1} \ge \ell_n + \frac{1}{2}\beta_{\min}$.

This restorative drift guarantees that the sequence (ℓ_n) is uniformly bounded. Thus, runs of consecutive A-bursts and B-bursts are strictly finite.
Bounded runs imply that each subsystem wins at least once every R bursts. Since a gadget's coordinate is multiplied by a factor strictly less than 1 upon winning, the total scale decays exponentially: $x_n + s_n \le C\rho^n$ for some $0 < \rho < 1$.
Consequently, the error terms are summable: $\sum_n |r_n| < \infty$.

Summing the increment equation gives:

$$\ell_n - \ell_0 = -\sum_{\substack{k<n,\\ A \text{ wins}}} \alpha_{i_k} + \sum_{\substack{k<n,\\ B \text{ wins}}} \beta_{j_k} + \sum_{k<n} r_k$$

Because ℓ_n is bounded and $\sum r_k$ converges, we obtain:

$$\sum_{\substack{k<n,\\ A \text{ wins}}} \alpha_{i_k} - \sum_{\substack{k<n,\\ B \text{ wins}}} \beta_{j_k} = O(1)$$

The phase index toggles each time a subsystem wins. Therefore, the counts of phase 0 and phase 1 among the wins differ by at most 1. Defining the averages $\bar{\alpha} = (\alpha_0 + \alpha_1)/2 = -\frac{1}{2}\log\lambda$ and $\bar{\beta} = (\beta_0 + \beta_1)/2 = -\frac{1}{2}\log\kappa$, we get:

$$\bar{\alpha} N_A(n) - \bar{\beta} N_B(n) = O(1)$$

Since $N_A(n) + N_B(n) = n$, the Cesàro frequency of A-wins is:

$$\frac{N_A(n)}{n} \to \frac{\bar{\beta}}{\bar{\alpha} + \bar{\beta}} = \frac{-\log\kappa}{-\log\lambda - \log\kappa}$$

Because $\log\lambda/\log\kappa \notin \mathbb{Q}$, this limit is irrational. Therefore, the burst-winner sequence is strictly aperiodic.
Since the phase pair updates as $(i, j) \mapsto (i \oplus 1, j)$ on an A-burst and $(i, j \oplus 1)$ on a B-burst, an eventually periodic phase pair would force an eventually periodic winner sequence. Thus, the phase pair (i_n, j_n) is also aperiodic.

8. No Cycle Convergence of the Distributions Define the four product cycle points $Q_{ij} = P_i^A \otimes P_j^B$ for $i, j \in \{0, 1\}$. These points are distinct.
Since $x_n, s_n \to 0$ and the cone sectors S_i^G are forward-invariant, the burst-boundary states approach the cycle points:

$$\operatorname{dist}(W_{5n}, Q_{i_n j_n}) \to 0$$

If the full AdaBoost trajectory W_t converged to an ℓ-cycle, the subsequence W_{5n} would also converge to a finite cycle. However, for all large n, W_{5n} enters four disjoint neighborhoods around the Q_{ij} dictated entirely by the label (i_n, j_n).
Convergence to a finite cycle would therefore force the phase pair (i_n, j_n) to be eventually periodic, resulting in a contradiction. Thus, AdaBoost on M_0 does not converge to any finite cycle for any initial point in the open product cone $S_0^A \times S_0^B$.

9. Returning to the True COLT Initialization The COLT problem requires the standard uniform initial distribution $D_1(i) = 1/m$.
Because $S_0^A \times S_0^B$ is a codimension-0 open set in the product manifold $\mathcal{P}$, rational points are dense within it. We can therefore choose a rational starting point:

$$W_0 = a_0 \otimes b_0 \in S_0^A \times S_0^B$$

Write the 20 coordinates of W_0 as $W_0(r) = n_r/L$ where $n_r \in \mathbb{N}$ and $\sum n_r = L$. Form the final matrix $\widetilde{M}$ by replacing row r of M_0 with n_r identical copies.

Run AdaBoost on $\widetilde{M}$ from the exact uniform start $D_1(i) = 1/L$. Duplicate rows receive identical multiplicative updates and remain equal forever. If π denotes aggregation over duplicate blocks, then $\pi(D_t) = W_t$, where W_t is exactly the orbit on M_0 started from W_0. The margin remains unchanged at $\gamma(\widetilde{M}) = 1/5$.
If the uniform-start orbit on $\widetilde{M}$ converged to a cycle, its aggregation W_t would too, leading to a contradiction.
Hence, standard exhaustive AdaBoost on the finite matrix $\widetilde{M}$, from a uniform initialization, does not converge to a cycle. ■

This initial proposal contained various imprecisions and other shortcomings, but was developed enough that we could push it toward a tighter argument. From this point, the work consisted primarily of prompting both Claude and GPT to critique GPT's solutions, then feeding these critiques to GPT in fresh conversations for revision. To avoid misleading GPT with spurious objections, we ran self-verification loops with both Claude and GPT in fresh conversations, iterating until it no longer produced false positives (correct aspects of the solution incorrectly flagged as errors). At the expository level, Claude was primarily used for resolving minor issues and adding technical detail to the text.

We observed that once GPT had identified the approach of constructing a counterexample from a set of sign-matrices that provably did not converge, it became far more autonomous and quickly arrived at the final structure of the solution presented here. This suggests that the model is currently capable of attacking a problem from many angles provided they lead to the same conclusion, but struggles to pivot between approaches with substantively different outcomes—in this case, a disproof via counterexample versus a proof of the conjecture. We also note that the model easily produced the code for the certificate. This is perhaps unsurprising given the coding ability of recent models, but we were nonetheless struck by how nearly it "one-shotted" the certificate.

Example of a Failed GPT Counterexample Attempt

I can close this along Path A, using the reduction in your item 8.
Let

$$L = \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & 1 \end{pmatrix}.$$

Call the first four rows (r_1, r_2, r_3, r_4), and the fifth row s.
The active four-row face $w_s = 0$ contains an exact period-10 AdaBoost orbit. The extra row s gives a passive transverse multiplier. The key point is that the passive multiplier and the leading intrinsic multiplier of the active return map have irrational log-ratio. Taking two factorized copies of this local system then gives an irrational rotation number, so the subsystem-selection itinerary is not eventually periodic.

1. An explicit local period-10 orbit On the face $w_s = 0$, write the active weights as (a, b, c, d) on (r_1, r_2, r_3, r_4).
For these four rows, exact line search has the usual "halve the two sign-groups" form. The

branch maps are

$$T_1(a,b,c,d) = \left(\frac{a}{2(1-b)}, \frac{1}{2}, \frac{c}{2(1-b)}, \frac{d}{2(1-b)}\right),$$

$$T_2(a,b,c,d) = \left(\frac{a}{2(a+b)}, \frac{b}{2(a+b)}, \frac{c}{2(c+d)}, \frac{d}{2(c+d)}\right),$$

$$T_3(a,b,c,d) = \left(\frac{a}{2(a+c)}, \frac{b}{2(b+d)}, \frac{c}{2(a+c)}, \frac{d}{2(b+d)}\right),$$

$$T_4(a,b,c,d) = \left(\frac{a}{2(a+d)}, \frac{b}{2(b+c)}, \frac{c}{2(b+c)}, \frac{d}{2(a+d)}\right).$$

Take the itinerary

$$1, 2, 1, 3, 4, 1, 2, 1, 4, 3.$$

Impose the half-cycle symmetry

$$T_4T_3T_1T_2T_1(a,b,c,d) = (a,b,d,c).$$

This forces

$$a = \tfrac{1}{2} - c, \qquad b = \tfrac{1}{2} - d,$$

and (c,d) must satisfy two polynomial equations. Eliminating d gives

$$p(c) = 2048c^7 - 2048c^6 - 640c^5 + 1328c^4 - 416c^3 + 4c^2 + 12c - 1 = 0.$$

Let c_* be the real root in $(0.2217, 0.2218)$:

$$c_* \approx 0.2217481715767779812786\ldots$$

Then

$$\begin{aligned} d_* &= -\frac{294912c_*^6 - 227840c_*^5 - 141056c_*^4 + 156768c_*^3 - 25836c_*^2 - 3660c_* + 719}{346} \\ &\approx 0.2364004668362263385034\ldots \end{aligned}$$

and therefore

$$\begin{aligned} a_* &= \tfrac{1}{2} - c_* \approx 0.2782518284232220187213\ldots, \\ b_* &= \tfrac{1}{2} - d_* \approx 0.2635995331637736614965\ldots. \end{aligned}$$

So

$$w_* = (a_*, b_*, c_*, d_*, 0)$$

is an exact point on a period-10 orbit for the 5-row local system L, with the stated itinerary. The chosen-edge values on one half-cycle are

$$\begin{aligned} &0.47280093367245\ldots, \quad 0.37785395440970\ldots, \quad 0.27423367563769\ldots, \\ &0.40494202576594\ldots, \quad 0.35010319148137\ldots \end{aligned}$$

and the competing edges are strictly smaller at every step, so there are no ties on this orbit.

2. The cumulative beta vector and the passive multiplier The four active rows satisfy

$$r_i \cdot (2,1,1,1) = 1 \qquad (i = 1,2,3,4).$$

Hence, by your item 6, the cumulative beta vector on this period-10 orbit has the form

$$\Gamma = g(2,1,1,1)$$

for some $g > 0$, and the common active value is $c = g$.
Let

$$\rho = e^{-g}.$$

Exact elimination from the orbit gives the minimal polynomial of ρ:

$$f(\rho) = \rho^7 - 9\rho^6 + 23\rho^5 - 3\rho^4 - 45\rho^3 + 13\rho^2 + 5\rho - 1 = 0,$$

with

$$\rho \approx 0.45153264872461054832291732265 9\ldots$$

Now choose the passive row

$$s = (1,1,-1,1),$$

which is exactly the fifth row of L. Since

$$s \cdot (2,1,1,1) = 3,$$

its passive exponent is

$$s \cdot \Gamma - c = (3-1)g = 2g.$$

So the passive multiplier is

$$\kappa = e^{-2g} = \rho^2.$$

Eliminating ρ gives the exact polynomial for κ:

$$h(\kappa) = \kappa^7 - 35\kappa^6 + 385\kappa^5 - 1835\kappa^4 + 2315\kappa^3 - 625\kappa^2 + 51\kappa - 1 = 0,$$

with

$$\kappa \approx 0.20388173286426254383101808231 2\ldots$$

Moreover, h is irreducible modulo 5, so it is the minimal polynomial of κ. Since h is monic of odd degree with constant term -1,

$$N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa) = (-1)^7(-1) = 1.$$

3. The leading intrinsic multiplier of the active return map Now look at the exact 10-step return map on the active 4-row face $w_s = 0$. In local coordinates (a,b,c), its Jacobian J at w_* has

$$\det J = 0,$$

and the two nonzero multipliers are the roots of

$$z^2 - \operatorname{tr}(J)\, z + e_2(J) = 0,$$

where $\mathrm{tr}(J)$ and $e_2(J)$ lie in $\mathbb{Q}(c_*)$.
Eliminating c_* against $p(c_*) = 0$ gives the exact degree-14 polynomial satisfied by the nonzero intrinsic multipliers:

$$\begin{aligned} g(z) = {} & 16z^{14} - 117232z^{13} - 8330040z^{12} - 3628852593z^{11} + 766183444916z^{10} \\ & - 37440753370248z^9 + 504085851461616z^8 + 57085439654130z^7 \\ & + 18299281845941356z^6 - 5676298472505780z^5 + 422743039128068z^4 \\ & - 4789675432569z^3 + 7468158864z^2 - 2899292z + 4. \end{aligned}$$

This polynomial is irreducible modulo 5, hence irreducible over $\mathbb{Q}$. Its larger root in $(0, 1)$ is

$$\lambda \approx 0.094492059489127164899022868216\ldots$$

This is the leading intrinsic stable multiplier of the period-10 return map on the active face. Because g is irreducible, the norm of λ is

$$N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda) = (-1)^{14}\frac{4}{16} = \frac{1}{4}.$$

So we now have two exact contraction factors:

$$\lambda \approx 0.09449205948912716, \qquad \kappa \approx 0.20388173286426254.$$

4. The log-ratio is irrational If $\log\lambda/\log\kappa$ were rational, then there would exist $m, n \in \mathbb{N}$ with

$$\lambda^m = \kappa^n.$$

Let $L = \mathbb{Q}(\lambda, \kappa)$. Taking absolute norms from L to $\mathbb{Q}$,

$$N_L(\lambda)^m = N_L(\kappa)^n.$$

But

$$N_L(\kappa) = N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa)^{[L:\mathbb{Q}(\kappa)]} = 1,$$

whereas

$$N_L(\lambda) = N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda)^{[L:\mathbb{Q}(\lambda)]} = \left(\frac{1}{4}\right)^{[L:\mathbb{Q}(\lambda)]} \neq 1.$$

Contradiction. Therefore

$$\frac{\log\lambda}{\log\kappa} \notin \mathbb{Q}.$$

Equivalently,

$$\frac{-\log\kappa}{(-\log\lambda) + (-\log\kappa)}$$

is irrational.
Numerically,

$$-\log\lambda \approx 2.3592394745829529, \qquad -\log\kappa \approx 1.5902151941140905,$$

so the rotation number is approximately

$$0.402640\ldots$$

and is irrational by the norm argument above.

5. The global counterexample Now take two independent copies of the same 5-row local system L. The global sign-pattern matrix is the explicit 25×8 matrix M with rows indexed by $(p, q) \in [5]^2$:

$$M_{(p,q),j} = L_{p,j} \quad (1 \le j \le 4), \qquad M_{(p,q),4+j} = L_{q,j} \quad (1 \le j \le 4).$$

So the row set is exactly

$$\mathrm{Rows}(M) = \mathrm{Rows}(L) \times \mathrm{Rows}(L) \subset \{\pm 1\}^8.$$

For the 10-step Poincaré return map of the local 5-row system at the period-10 orbit, the spectrum is

$$\{\kappa, \lambda, \lambda_2, 0\}$$

with

$$0 < \lambda_2 < \lambda < \kappa < 1.$$

Because κ and λ are simple and spectrally separated, standard invariant-manifold theory gives one-dimensional analytic invariant manifolds W^κ and W^λ tangent to the κ- and λ-eigendirections. By Koenigs linearization, on those manifolds the return map is analytically conjugate to

$$u \mapsto \kappa u, \qquad v \mapsto \lambda v.$$

Take one copy initialized on $W^\lambda \setminus \mathcal{O}$ and the other on $W^\kappa \setminus \mathcal{O}$, both sufficiently close to the local cycle. Since both copies are built from the same local orbit, the baseline best-edge sequence is identical in the two factors. The competition exponents are therefore

$$\alpha = -\log \lambda, \qquad \beta = -\log \kappa,$$

with $\alpha/\beta \notin \mathbb{Q}$.
Now invoke your item 8: the induced competition map on the phase circle is rigid rotation with irrational rotation number. Hence the subsystem-selection itinerary is Sturmian and **not eventually periodic**. Choosing the initial phase off the countable boundary orbit keeps all comparisons strict, so the no-tie condition holds for the whole orbit.
Therefore the resulting AdaBoost trajectory for this explicit M does **not** phasewise converge to a finite cycle.

Conclusion. This gives an explicit disproof of "AdaBoost always cycles" in the phasewise sense.
The simplest passive/passive construction still fails for the reason you identified: all passive exponents are rational multiples of the common scalar g. The actual counterexample uses one passive multiplier and one intrinsic local multiplier of the same visible cycle. Their norms are different (1 versus 1/4), which forces irrational log-ratio and hence irrational rotation number in the asynchronous product.
So the conjecture is false.

While Claude could not reliably determine the correctness of GPT's attempts at producing counterexamples, GPT itself could—through both follow-up prompts in the same conversation and queries in fresh conversations.

B.1 Role of the human

While the models drove the majority of the reasoning process, the human authors guided the overall strategy using carefully crafted prompts and by prioritizing certain directions. In particular, since the model could recognize the open status of the problem, it often refused to solve the problem–therefore, adding certain directions was crucial to making progress on the problem. Additionally, it took roughly two weeks and slightly less than 100 prompts with GPT-5.4 Pro to produce the final solution. Since each GPT-5.4 Pro (extended thinking) query could take up to two hours to complete, the selection of which directions to pursue and which to abandon was important from a resources perspective. One such example is the explicit instruction to GPT to pursue the disproof approach, given its numerous failures in attempting a proof.

We reviewed the final solution written by the models along with the certificate code. This component is still critical to the research process, as both Claude and GPT frequently missed gaps, errors, and ambiguities in the solution. While we found that multiple fresh conversations with the models and carefully crafted prompts (using language such as "be very careful in your review, as ambiguity or imprecision in the solution are unacceptable") generally led to the highest-quality reviews from the models, there were still several areas in the solution that required additional elaboration and clarification that were only identified by human experts. That said, the speed and sophistication with which the top-performing models reasoned through and documented their findings suggest that their capabilities may already be comparable to those of human experts in the field.

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